
Q/R/M/C: A Stage-Decomposition Measurement Framework for Memory-Based Navigation

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Abstract

Setting. Success-only metrics collapse qualitatively different memory-based navigation failures into a single scalar, hiding *which stage* of the retrieval pipeline is responsible. This paper contributes an *evaluation protocol*: the Q/R/M/C stage decomposition (**Q**uery formation, **R**etrieval satisfaction, target **M**aterialization, **C**ompletion after retrieval), operationalised so that the four conditional rates are computable directly from standard trajectory logs (Definition 1). The arithmetic is the chain rule, deliberately so — just as decomposing accuracy into precision and recall added no mathematics but changed what the field could *see*, the value here is identifiability from logs and what the decomposition reveals, not a theorem.

Single-agent measurement. On a 10-seed MiniGrid benchmark (FourRooms, LavaGapS7) and a BabyAI portability check, the framework localises bottlenecks: hazard recovery is retrieval-limited (oracle retrieval lifts success $0.39 \rightarrow 0.60$), goal-region rejoin is downstream-limited. Log inspection refines the attribution: candidate ranking is near-optimal (96.9%) while candidate generation fails at 91% of decision points — a memory-accumulation, not a ranking, limit.

Multi-agent measurement. On a 35,640-run multi-agent sweep over four communication architectures (independent, shared-bus, central-aggregator, peer-broadcast at eight cadences $K \in \{1, 2, 4, 8, 16, 32, 48, 64\}$, 40 seeds), the bottleneck-shift between the M and C links appears directly as a negative within-cadence correlation peaking at $K=4$ (Pearson -0.61); mean time-to-success has an interior minimum at $K=8$ (robust in location but shallow in margin — not resolved from the adjacent fast cadences). Both slope signs (Empirical Claim 1) hold as moderate, cluster-robust effect sizes (Spearman -0.53 and $+0.43$ vs. K , 95% CIs $[-0.59, -0.47]$ and $[+0.37, +0.50]$ over 81 configuration cells; Appendix F.3). Non-claims (notably on the conditional product Φ) are consolidated in one place (§1, Scope and non-claims).

Diagnostic use. The same four rates also expose why learned policies with implicit communication can be competitive on success while collapsing Q/R/M/C observability: their communication vectors are never informatively empty and they lack an explicit target-lock interface. On two local LLM agents (qwen3:4b, llama3.1 on TextNav), the same logger localizes a completion-limited failure: under a tight step budget both agents form the query, retrieve, and lock the target ($Q^*=R^*=M^*=1$) yet none complete ($C^*=0$), so an otherwise opaque success collapse is resolved into a specific failing stage rather than a leaderboard number. Thus the contribution is a measurement protocol for making memory failures comparable across symbolic, learned, LLM, single-agent, and multi-agent systems.

Scope. The primary experiments are in grid-world substrates (MiniGrid, BabyAI, a custom 12×10 grid). The contribution is the diagnostic framework and its validation; a preliminary continuous-state check on VMAS reproduces both slope signs directionally (Appendix F.7), and a full continuous/3D validation with cluster-robust CIs (Habitat, ProcTHOR) is future work.

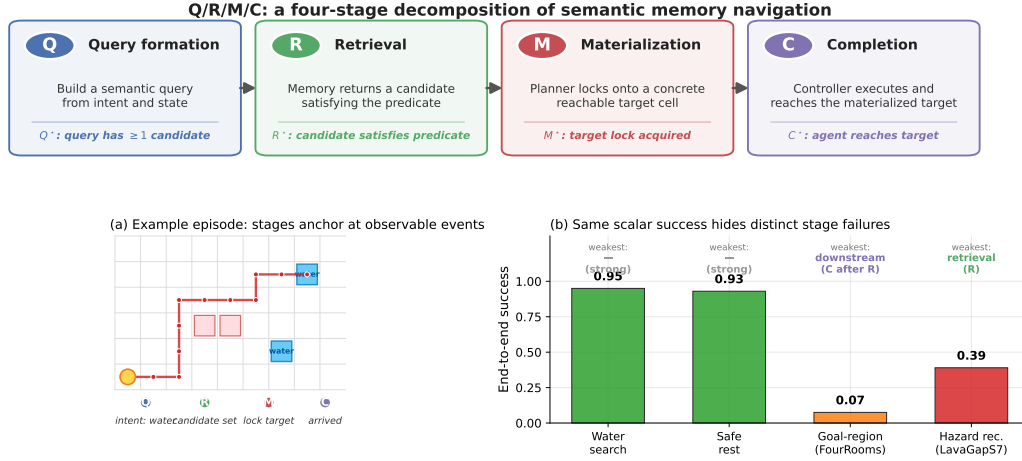


Figure 1: *Q/R/M/C* decomposes one semantic-navigation episode into four observable stages. **Top:** the four stages and their event-level definitions. **(a)** A successful water-search episode anchors at one event per stage, producing logged starred events Q^* , R^* , M^* , C^* . **(b)** The same scalar “end-to-end success” summary masks qualitatively different failures on the four single-agent task families: water and rest are strong; goal-region rejoin (FourRooms) is downstream-limited; hazard recovery (LavaGapS7) is retrieval-limited (oracle interventions, §3.4).

1 Introduction

Classical navigation benchmarks often assume that the agent is given an explicit destination coordinate or a dense task reward. Many practical navigation problems are different: the agent must reach a place of a certain *type*, such as a safe rest area, a water-like resource, a post-hazard recovery exit, or a goal-associated region, specified by a semantic pattern rather than a fixed (x, y) waypoint. The place must therefore be *remembered and retrieved by meaning*. This setting opens a methodological question: can we measure *where* memory-based semantic navigation breaks down, not only *whether* it succeeds?

We answer this question with a four-stage diagnostic framework that applies to both single agents and multi-agent collectives. The decomposition is *Q/R/M/C*: **Q**uery formation, **R**etrieval satisfaction, **M**aterialization, and **C**ompletion after retrieval. The decomposition factorizes per-episode semantic-path completion and, under a conditional stage-Markov property, its factors are estimable from trajectory logs (Definition 1). The rest of the paper develops a single argument in three parts.

Single agent. Stage attribution is non-trivial. Success-only metrics collapse qualitatively different failures into a single scalar. In a 10-seed MiniGrid benchmark, water and rest tasks are strong end-to-end (0.95, 0.93 success), but goal-region rejoin and hazard recovery are substantially harder, and for *different reasons*. Oracle-stage interventions show that hazard recovery is retrieval-limited (oracle retrieval lifts success $0.39 \rightarrow 0.60$), while goal-region rejoin is downstream-limited (oracle retrieval rescues semantic-path completion without changing end-to-end success). Inspection of the framework’s own logs then sharpens the attribution: the retrieval bottleneck is not candidate *ranking* (precision 96.9% when candidates exist) but candidate *generation*: the memory contains no eligible symbols at 91% of decision points. The single-agent bottleneck is a *memory-accumulation* limit: evidence is observed but never consolidated into queryable structure.

Multi-agent. Consolidation acquires a rate. When N agents share memory through a broadcast protocol with cadence K , the factorization extends with enlarged conditioning sets (Definition 2), and K becomes a structural axis: it is, mechanically, a knob on *how fast private memories consolidate into a collective one*. A slope-level claim (Empirical Claim 1) predicts a bottleneck shift: as K decreases, $P(M^*|R^*)$ rises (fresher peer memory enables reliable commitment) and $P(C^*|M^*)$ falls (peers race for the same targets). Both slope signs are confirmed on a 35,640-run sweep as moderate, cluster-robust effect sizes (Appendix F.3), the shift is directly visible as a negative within-cadence

71 correlation between the two links, and the resulting trade-off produces an interior optimum of mean
72 time-to-success at $K = 8$: consolidation has a non-trivial optimal rate.

73 **Diagnostic synthesis.** Both regimes point at the same measurement object. The single-agent diagnosis
74 says that evidence consolidation into symbols can be the binding constraint; the multi-agent diagnosis
75 says that the rate and cost of sharing consolidated memory determine the optimum. Q/R/M/C is the
76 common instrument: candidate systems can be compared by how they move $P(R^*|Q^*)$, $P(M^*|R^*)$,
77 and $P(C^*|M^*)$, even when their end-to-end success rates look similar.

78 **Scope and non-claims (consolidated).** All caveats of this paper in one place; the running text
no longer repeats them. (i) The decomposition’s consistency argument is the chain rule plus
frequency estimation — no mathematical novelty is claimed; the contribution is the logging
discipline and identifiability (Definition 1–2). (ii) Empirical Claim 1 is a verified slope-level
observation on the conditional rates, not a theorem. (iii) We make no claim about the shape
of the conditional product Φ ; in our data it is monotone in K , and the interior optimum
lives on mean time-to-success (§4.5). (iv) Primary substrates are grid worlds; VMAS is a
directional continuous-state check, and the LLM study is a small controlled one (two local
models, TextNav). (v) Multi-agent place identity assumes a shared coordinate frame (§5). A
stage-level operationalisation is testable on specific summaries and can be *wrong* on others
— (iii) is an instance we report — which is what separates a measurement protocol from a
stage-agnostic guess.

79 **Why this framing is timely.** Embodied and goal-conditioned agents increasingly retrieve places,
80 objects, or tools by meaning (Andrychowicz et al., 2017; Ghosh et al., 2021; Savva et al., 2019).
81 LLM-agent systems lean on explicit memory layers but evaluate mostly end-to-end (Wang et al., 2023;
82 Shinn et al., 2023; Packer et al., 2023). Multi-agent reinforcement learning likewise increasingly
83 involves communicating populations (Lowe et al., 2017; Foerster et al., 2018), but the question
84 “which agent should know what, and when?” lacks a unified diagnostic vocabulary. Our framework
85 supplies one.

86 **Contributions.** This is a methodology paper. We are explicit about what is bookkeeping vs. empirical
87 claim vs. instantiated artefact.

- 88 1. **Measurement protocol (single-agent).** The Q/R/M/C decomposition is operationalised so
89 that the conditional rates are estimable from trajectory logs (Definition 1); the contribution
90 is the *logging discipline* that makes the decomposition usable on standard navigation traces
91 (non-claims: see box above).
- 92 2. **Measurement protocol (multi-agent).** An extension to N communicating agents with two
93 coupling channels (memory $\mathcal{M}_{-i}^{(K)}$, occupancy $\mathcal{O}_{-i}$) under a peer-broadcast protocol with
94 cadence K (Definition 2).
- 95 3. **Empirical Claim 1 (bottleneck shift).** On a 35,640-run multi-agent sweep, the two stylised
96 slope conditions **(F)** and **(O)** are both confirmed as moderate, cluster-robust effect sizes (Spear-
97 man -0.53 and $+0.43$ vs. K , 95% CIs excluding zero over 81 configuration cells; Appendix F.3;
98 full benchmark in §4.5), with a provenance-level mechanism (§4.5).
- 99 4. **System and benchmarks.** A symbolic-semantic memory stack with four communication
100 architectures (independent, shared-bus, central-aggregator, peer-to-peer broadcast); single-agent
101 oracle benchmark (10-seed MiniGrid, BabyAI portability); 35,640-run multi-agent sweep at
102 $N \in \{3, 5, 8\} + 8,640$ -run scale-up at $N = 12$; 100-run portability sweep on a standard-MiniGrid
103 wrapper.
- 104 5. **Diagnostic comparison across architectures.** The same logging contract separates symbolic
105 memory systems from learned policies with implicit communication: the latter can be competi-
106 tive on success while collapsing Q/R/M/C stage observability. This is an evaluation result, not a
107 claim that one architecture class should always dominate.
- 108 6. **Scope and honesty.** The primary experiments are in grid-world substrates. We treat that
109 as a methodology-paper choice and state it explicitly (§5); a preliminary VMAS check on a
110 continuous substrate is included (Appendix F.7), and full continuous/3D validation (Habitat,
111 ProcTHOR, MineDojo) is future work.

2 The Q/R/M/C decomposition

2.1 Problem formulation and notation

The agent operates in a partially observable navigation environment. At each step it receives an observation o_t , maintains a memory state $\mathcal{M}_t$, and forms a query q_t encoding the current task intent. A retrieval step returns a candidate set of places matching q_t from $\mathcal{M}_t$; a chosen candidate is materialized into an actionable target; a local controller executes actions until either the target is reached or the episode times out. The target is not a coordinate but a semantic pattern, so success is a function of the chosen candidate’s semantic profile, not of any pre-supplied coordinate.

Habitat, AI2-THOR, and ProcTHOR made visually rich navigation evaluation standard (Savva et al., 2019; Kolve et al., 2017; Deitke et al., 2022). We deliberately focus on MiniGrid-class environments to obtain clean and reproducible stage measurements, and use BabyAI only as a portability check. We do not claim to solve LLM-agent memory here; rather, we argue that the same Q/R/M/C decomposition is a useful template for evaluating memory-grounded agent decisions beyond navigation.

The central empirical question is therefore not just whether the agent succeeds, but *which stage fails first* when it does not. Two related measurement layers are used. **Operational rates** count, per query attempt, whether the query yielded a non-empty candidate set, whether the chosen candidate satisfied the semantic predicate, and whether it was materialized into an actionable target; *completion after retrieval* is an episode-level indicator. **Nested episode-level events** $Q^* \Rightarrow R^* \Rightarrow M^* \Rightarrow C^*$ are used in the factorization below and audited in §3.2. Full operational definitions are given in Appendix C.

Notation. The four stages are Q, R, M, C with episode-level starred events $Q^*, R^*, M^*, C^* \in \{0, 1\}$, nested by construction. $P(\cdot)$ denotes the population probability; $\hat{P}(\cdot)$ its empirical-frequency estimator. **The letter M always refers to the Materialization stage; the number of targets in the multi-agent environment is written T (not M , constraint $T \leq N$, “scarcity” = $N > T$).** Other symbols: $\mathcal{C}_K$ peer-broadcast protocol with cadence K , $\mathcal{M}_{-i}^{(K)}$ peer memory snapshots, $\mathcal{O}_{-i}$ peer occupancy, $\nu(\cdot)$ freshness functional, $\Phi(K) = P(M^* | R^*; K) \cdot P(C^* | M^*; K)$, t_{succ} mean time-to-success. Full symbol table in Appendix D.1.

2.2 Single-agent factorization

Let Y denote overall episode success, and let Q^*, R^*, M^*, C^* denote the nested episode-level semantic-path events. The framework factorizes *semantic-path completion* rather than raw task success:

$$Q^* \rightarrow R^* \rightarrow M^* \rightarrow C^*.$$

For any query q and memory state $\mathcal{M}$,

$$P(C^* = 1 \mid q, \mathcal{M}) = P(Q^* = 1 \mid q, \mathcal{M}) P(R^* = 1 \mid Q^* = 1, q, \mathcal{M}) \\ \cdot P(M^* = 1 \mid R^* = 1, q, \mathcal{M}) P(C^* = 1 \mid M^* = 1, q, \mathcal{M}).$$

Overall task success Y can exceed C^* because some episodes succeed without traversing the full semantic retrieval path.

Assumption 1 (conditional stage-Markov property). Conditioned on the current stage being successful, the probability of the next stage depends only on the current successful stage, the current query, and the current memory state. Formally,

$$P(R^* \mid Q^*, q, \mathcal{M}, \text{history}) = P(R^* \mid Q^*, q, \mathcal{M}),$$

and analogously for M^* and C^* .

Assumption 2 (positivity). Each conditioning event used above has non-zero probability on the support of the benchmark.

Definition 1 (single-agent stage estimability). Under Assumptions 1–2, the four conditional rates in the factorization above are well-defined population quantities and the empirical-frequency estimators are consistent estimators of them; their product converges to semantic-path completion.

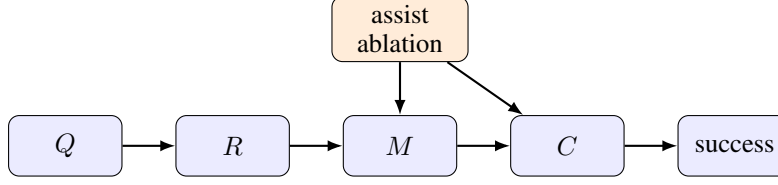


Figure 2: Stage-structured interpretation of the Q/R/M/C framework. Assist on/off ablations target intermediate mechanisms; in the multi-agent extension the cadence protocol $\mathcal{C}_K$ acts on the $R \rightarrow M$ and $M \rightarrow C$ edges.

The argument is elementary — the chain-rule factorization plus consistency of frequency estimators on Bernoulli events — and the value of stating it is operational, not mathematical: it pins down the *logging discipline* (which events, conditioned on what) under which the decomposition is readable from standard trajectory logs, and it is the contract that every experiment in this paper audits against (§3.2). Full statement and elementary argument in Appendix A.

When Assumption 1 may fail. Assumption 1 holds tightly here because q and $\mathcal{M}$ are explicitly logged at every stage event. Two practically relevant violations remain: adaptive in-episode query rewrites not surfaced to the logger, and hidden between-stage memory updates. In those cases the factorization should be read as a projection onto observable traces rather than as a full structural identification claim. Appendix F.6 quantifies the estimator bias under a controlled violation of this kind: it is negligible for mild unlogged couplings and one-signed (an upward bias on $P(C^*|M^*)$), which bounds the framework’s exposure when it is applied to learned or LLM agents.

Mechanism-level reading. The same decomposition admits a stage-structured mechanism interpretation. Figure 2 treats stage outcomes as a directed acyclic graph. Under this interpretation, assist on/off ablations are mechanism-targeted perturbations of materialization and post-retrieval execution rather than undifferentiated sensitivity analyses; in the multi-agent setting the same graph is replicated per agent and connected through the cadence protocol $\mathcal{C}_K$, which acts on $\mathcal{M}_{-i}^{(K)}$ at the $R \rightarrow M$ edge and on $\mathcal{O}_{-i}$ at the $M \rightarrow C$ edge — exactly the two edges where Empirical Claim 1 will predict the slope shift.

Relation to prior work. Closest are (i) cognitive-map and semantic-map frameworks (Tolman, 1948; O’Keefe and Nadel, 1978; Kuipers, 2000; Gupta et al., 2017; Savinov et al., 2018), (ii) goal-conditioned and successor-representation RL (Andrychowicz et al., 2017; Ghosh et al., 2021; Dayan, 1993; Stachenfeld et al., 2017; Momennejad et al., 2017), (iii) multi-agent communication (Lowe et al., 2017; Foerster et al., 2018; Sukhbaatar et al., 2016; Foerster et al., 2016) and gossip/consensus (Boyd et al., 2006; Xiao et al., 2007), and (iv) LLM-agent memory layers evaluated end-to-end (Wang et al., 2023; Shinn et al., 2023; Packer et al., 2023). Our framework is orthogonal to all four: it *diagnoses* where in the Q/R/M/C chain a given architecture succeeds or fails and lifts the cadence axis from network protocols to a cognitive-architecture parameter (full discussion in Appendix F.9).

2.3 Event semantics, bounded rationality, and an order-theoretic reading

The four starred events anchor at concrete logged triggers. We state the triggers once, because every empirical claim in the paper is a claim about them (per-regime formal definitions in Appendix C), and we give each stage its reading in terms of the resource it bounds.

Q^* (intent issued). The planner emits a semantic intent $q = (q^{\text{req}}, q^{\text{pref}}, q^{\text{pen}})$: small sets of required, preferred, and penalty tags. The episode-level event fires on emission, so it is saturated whenever the task is semantic ($\hat{P}(Q^*)=1.00$ in Table 1); the multi-agent per-tick logger additionally requires a non-empty candidate set (Appendix C.1), a bookkeeping difference we record rather than hide. Resource bound: attention — the intent is a few tags, not the full state.

R^* (memory answers). Retrieval evaluates q against the memory and returns a candidate set; the event fires when the set contains a candidate satisfying the task predicate (within ϵ of a true target in the multi-agent logger). The 91% empty-candidate-set failures of §3.5 are R -side events: no

195 stored place clears the required-tag threshold. Resource bound: memory — retrieval can only
 196 surface what consolidation has stored.

197 **M^* (target lock).** The planner materializes one candidate into a concrete actionable cell. Candidacy
 198 itself is threshold-gated (a place enters the candidate set only if its score clears the planner’s gate,
 199 Appendix D), and the lock commits to a single candidate once per query rather than re-optimizing
 200 against the full memory at every tick. The event fires on lock acquisition (within ϵ of a true target
 201 in the multi-agent logger). Resource bound: deliberation.

202 **C^* (closed-loop completion).** The local controller executes until the locked cell is reached or the
 203 episode times out; the event fires on arrival. C is the only stage whose failure modes are extra-
 204 semantic: controller competence and, in the multi-agent regime, occupancy $\mathcal{O}_{-i}$. Resource bound:
 205 control, and in a collective, shared targets.

206 **Why a deterministic strategy: bounded and ecological rationality.** A natural objection is that the
 207 semantic stack is a fixed, deterministic strategy rather than a learned policy. We treat the determinism
 208 as principled and name it: the pipeline is a satisficing agent in Simon’s sense (Simon, 1955, 1956).
 209 Candidacy is threshold-gated instead of exhaustively scored, the target commitment is made once
 210 instead of re-optimized at every tick, and the intent is a few tags instead of the full state. Bounded
 211 rationality is also what makes the four failure loci observable: each stage is a distinct resource bound
 212 (attention, memory, deliberation, control), and Q/R/M/C is the audit trail of where the bound binds.
 213 The learned baselines make the converse point empirically: policies without an explicit query/lock
 214 interface collapse the stages even at competitive success (BC in §3.3; CommNet PPO in §4.5), so
 215 observability is traded away exactly where end-to-end optimization is traded in. That the *same*
 216 heuristic is strong on water and rest, retrieval-starved on hazard recovery, and downstream-limited on
 217 goal-region (Figure 1b) is the ecological-rationality point that heuristic success is a property of the
 218 heuristic–environment pair (Simon, 1956; Gigerenzer and Goldstein, 1996); the framework is the
 219 instrument that locates the mismatch. The same lens reads the Φ -versus- t_{succ} split of §4.5: the chain
 220 probability Φ ignores the cost of traversing the chain, so under a resource-rational reading (Lieder
 221 and Griffiths, 2020) an optimum should be sought on a cost-bearing metric; that is where it appears
 222 ($K=8$ on mean time-to-success) and not on Φ .

223 **An order-theoretic reading (deferred to Appendix A.4).** The query–retrieval interface carries a
 224 standard order-theoretic structure: places and tags form a formal context whose Galois connection
 225 realizes the memory’s concept lattice, so retrieval fails empty exactly when the queried intent’s extent
 226 is empty — “absence of evidence,” the 91%/96.9% regime of §3.5, and the regime where growing
 227 the context can only grow extents. We use this as a naming device, not a contribution; the statement,
 228 its four-line argument, and its scope limits are in Appendix A.4.

229 **A dynamic (options / H-MDP) reading (deferred to Appendix B).** The stages also admit a
 230 temporally-extended-action reading: each query instantiates an option $o = \langle \mathcal{I}, \pi, \beta \rangle$ (Sutton et al.,
 231 1999) whose internal structure Q/R/M/C audits edge-for-edge (initiation $\rightarrow$ high-level policy over
 232 options $\rightarrow$ execution-to-termination), and the semi-MDP execution duration τ explains structurally
 233 why the interior optimum appears on t_{succ} but not on the τ -blind chain probability Φ . Like the
 234 order-theoretic reading, this is an interpretive lens, not a contribution; the full development, the
 235 belief–desire–intention gloss (shared with the collective-memory companion), and the multi-agent
 236 option-collision account of the bottleneck shift are in Appendix B.

237 3 Single agent: stage diagnostics localize failure

238 3.1 System, tasks, and protocol

239 **Memory stack.** The system uses a graph substrate with a semantic retrieval interface on top:
 240 observations o_t are tokenized into symbolic scene tokens and semantic tags that update a memory
 241 of place records, transitions, episodic traces, and concept/relation records. Planner queries return
 242 candidates by a weighted log-score over required, preferred, and penalty tags. The retrieval machinery
 243 is reused across all tasks and architectures; full memory schema, update rules, and scoring formula
 244 are in Appendix D.

Table 1: Stage-factor consistency audit for the best semantic method in each task family. The product of nested conditional estimators matches semantic-path completion; overall success can be higher because some episodes succeed without traversing the full semantic retrieval path.

Task	Method	Overall	SP-comp	$\hat{P}(Q^*)$	$\hat{P}(R^* Q^*)$	$\hat{P}(M^* R^*)$	$\hat{P}(C^* M^*)$	Product
Water	water-cp	0.95	0.70	1.00	0.98	0.74	0.97	0.70
Rest	rest-s2	0.93	0.73	1.00	1.00	0.80	0.91	0.73
Goal region	goal-n1	0.54	0.00	1.00	0.54	0.01	0.00	0.00
Hazard recovery	hazard-s7	0.40	0.16	1.00	0.56	0.58	0.50	0.16

Tasks, baselines, protocol. Four single-agent task families are evaluated (water, safe-rest, goal-region on Empty-6x6 / FourRooms (Sutton et al., 1999), hazard recovery on LavaGapS7), 10 seeds $\times$ 8 episodes each, $\times$ 2 assist modes. Semantic stack (node, concept, state-conditioned variants) is compared against four baselines: random, graph-only, SPTM-like, learned BC. Reported: end-to-end success, steps, four operational stage rates, assist deltas, weakest stage per task and per method. All means are seed-aggregated; 95% bootstrap CIs use $B=5,000$ resamples (Efron, 1979); assist comparisons use paired Wilcoxon signed-rank tests (Wilcoxon, 1945) with Holm-Bonferroni correction (Holm, 1979) (Appendix G). For the two hardest tasks (FourRooms, LavaGapS7), four oracle regimes (*base*, *oracle R/M/C*) are run — each replaces exactly one stage while leaving the rest of the pipeline unchanged. BabyAI portability check in Appendix E. All single-agent experiments ran on a 10-core ARM CPU, 32 GB RAM (CPU-only); oracle sweep < 1 min wall-clock.

LLM diagnostic block. To check that Q/R/M/C is not tied to the symbolic planner implementation, we also run a fixed TextNav substrate with an LLM-driven observe/query/retrieve/decide loop. The TextNav task, prompts, parser contract, and event definitions are held fixed while only the local Ollama model changes; the backend uses model-compatible completion settings when an Ollama chat template otherwise hides short structured answers in a separate thinking field. This block is a controlled step-budget study (Table 3): sweeping the completion budget drives both LLM agents into a failure that Q/R/M/C localizes to the C stage while $Q/R/M$ stay saturated, exhibiting the framework’s diagnostic property on real language agents. Reproducing full ReAct/Reflexion/Voyager/MemGPT-style agents is a separate systems contribution; the comparison to those systems is a coverage comparison in Appendix F.10.

3.2 Stage-estimator validation

Before the main benchmark, the factorization is validated. On a synthetic four-stage process with known probabilities the nested-estimator product tracks empirical semantic-path completion across sample sizes. On the real benchmark, the same product matches semantic-path completion rather than overall success (Table 1) — exactly the contract of Definition 1.

3.3 Success-only summary and what it hides

Water and rest are strong end-to-end tasks (0.95 and 0.93 success; Figure 1b). Goal-region reaches 0.54 success at the task-family level — an average that pools a solved environment (Empty-6x6, 1.00) with a hard one (FourRooms, 0.075; §3.4), so the family number is bimodal rather than a uniform mid-difficulty — but the bottleneck sits downstream of candidate selection (§3.4). Hazard recovery is the hardest retrieval-sensitive task at 0.40 success, with retrieval satisfaction as the weakest stage. Under success-only evaluation the two hard families look like the same kind of weakness; the stage view will show they are opposites. Per-task operational rates with 95% CIs are in Table 4 (Appendix D.3).

Baseline comparison. Table 2 compares the semantic stack against random, graph-only, SPTM-like, and learned behavior-cloned baselines. The learned baseline dominates the easiest resource tasks (1.00 on water and rest) and hazard recovery (0.74), while the semantic stack remains strongest on the relation-heavy goal-region family (0.54). The relevant contrast for this paper, however, is not the success column: the semantic stack is the only method family whose decisions expose interpretable Q/R/M/C structure. On the BC baseline the stage events are degenerate for the same structural reason

Table 2: Baseline comparison. The learned behavior-cloned baseline dominates the easiest resource tasks and hazard recovery, while the semantic stack remains strongest on the relation-heavy goal-region family and remains the only family with interpretable Q/R/M/C structure.

Task	Random	Graph-only	SPTM-like	Learned	Best semantic
Water	0.85	0.90	0.90	1.00	0.95
Rest	0.85	0.90	0.90	1.00	0.93
Goal region	0.41	0.49	0.52	0.50	0.54
Hazard recovery	0.08	0.39	0.00	0.74	0.40

as on the multi-agent learned baseline analysed in §4.5 — a policy without an explicit query/lock interface cannot emit separable stage events, so the diagnostic decomposition has nothing to attach to. End-to-end learning and stage diagnostics answer different questions; this paper is about the second.

3.4 Oracle-stage interventions localize the bottleneck

On goal-region the semantic variants solve Empty-6x6 (1.00 success) but collapse on FourRooms (0.075). For hazard recovery (LavaGapS7), base success 0.39 rises only to 0.43 under oracle controller, but jumps to 0.59 under oracle materialization and 0.60 under oracle retrieval; goal-region rejoin is downstream-limited (oracle R lifts semantic-path completion to 0.08 without moving end-to-end success). The per-task attribution as a stacked waterfall is in Figure 5 (Appendix D.3). Two tasks identical under success-only evaluation fail at opposite ends of the Q/R/M/C chain.

3.5 The logs refine the diagnosis: a memory-accumulation limit

Inspecting the framework’s own logs sharpens the retrieval attribution. On the 10-seed hazard-recovery run, 349 retrieval calls were logged: only 32 (9.2%) returned at least one candidate, and of these 31 (96.9% precision) selected a semantically satisfied candidate. The bottleneck is therefore not candidate *ranking* but candidate *generation* — the symbolic memory does not contain hazard-recovery-tagged nodes at 91% of decision points. A retrained 2-layer candidate-generation MLP trained on per-state features (test accuracy 0.59 vs majority-class 0.81; Appendix H) confirms that instantaneous state features cannot substitute for the missing structure. The framework’s “retrieval-limited” attribution thus refines to a *memory-accumulation* limit: candidate generation cannot be solved from state features alone, and the binding constraint is how trajectory evidence is consolidated into persistent symbolic structure. This attribution is not an artefact of the required-tag threshold θ : sweeping θ across the full range leaves the empty-candidate-set rate invariant (Appendix F.5), because the empties are absence of any stored candidate, not sub-threshold filtering. This sets up the multi-agent section: when consolidation becomes a shared process across N agents, it acquires a measurable *rate*.

4 Multi-agent: memory as a shared resource

4.1 Four memory architectures and the cadence axis

Four communication architectures are instantiated; they share the single-agent retrieval substrate but differ in how memory is distributed across agents (Figure 3).

Independent. Each agent maintains a private event store and concept graph. No inter-agent communication mechanism exists. Lower-bound baseline.

Shared bus. All agents publish observations to a single *collective memory* event bus. One shared concept graph is rebuilt from the union of events. Maximally centralized.

Central aggregator (ConsensusLayer). Each agent has its own private concept graph; a central layer pulls snapshots and computes a trust-weighted merge, producing a single *consensus report* consumed by all agents.

Peer-to-peer broadcast (cadence K). Each agent has its own concept graph and its own merged view. Every K ticks each agent broadcasts a snapshot to a passive bus; each agent independently

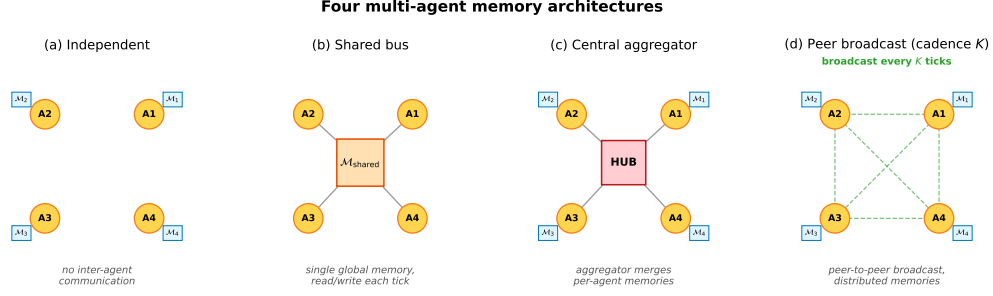


Figure 3: Four multi-agent memory architectures. **(a) Independent:** per-agent private memory, no exchange. **(b) Shared bus:** one global memory, every agent reads/writes each tick. **(c) Central aggregator:** per-agent memories plus a hub that merges them into a consensus report. **(d) Peer-to-peer broadcast:** per-agent memories with pairwise snapshot exchange every K ticks (the cadence axis of Empirical Claim 1).

merges its own snapshot with the latest snapshot received from each peer. Different agents may hold different merged views; there is no central report.

The first three correspond to the three regimes in the multi-agent communication literature: no exchange, CTDE-style aggregation, and decentralized peer communication. The fourth is the architecture for which Definition 2 and Empirical Claim 1 are directly applicable, since the cadence K is a tunable parameter rather than a fixed protocol. Mechanically, K is the rate at which private memories consolidate into a collective view, which is why it is the structural axis of this section.

4.2 Multi-agent factorization

Extend the factorization to a population of N agents that share information through a communication protocol $\mathcal{C}_K$ parameterized by broadcast cadence K . Each agent i has its own memory state $\mathcal{M}_i$ and issues its own query q_i . Let $\mathcal{M}_i^{(K)}$ denote the projection of agent i 's memory at the moment it queries, after absorbing every peer message received under cadence K . Two coupling channels are separated: (i) memory coupling $\mathcal{M}_{-i}^{(K)}$, the latest peer memory snapshots accessible to agent i ; (ii) occupancy coupling $\mathcal{O}_{-i}$, the set of targets that other agents have already claimed by the time i acts.

Definition 2 (multi-agent stage estimability). For each agent i ,

$$P(C_i^* \mid q_i, \mathcal{M}_i, \mathcal{C}_K) = P(Q_i^* \mid q_i, \mathcal{M}_i) P(R_i^* \mid Q_i^*, q_i, \mathcal{M}_i, \mathcal{M}_{-i}^{(K)}) \\ \cdot P(M_i^* \mid R_i^*, q_i, \mathcal{M}_i, \mathcal{M}_{-i}^{(K)}) P(C_i^* \mid M_i^*, q_i, \mathcal{O}_{-i}).$$

Under a multi-agent stage-Markov assumption (Appendix C), the four factors are estimable from per-agent trajectory logs and the empirical-frequency estimators are consistent.

When Definition 2 may fail. The single-agent stage-Markov property absorbs trajectory history into $(q, \mathcal{M})$. In the multi-agent setting, the analogous absorption requires that dependence on past peer decisions enters only through $\mathcal{M}_{-i}^{(K)}$ at the M stage and through $\mathcal{O}_{-i}$ at the C stage. This step is non-trivial: $\mathcal{O}_{-i}$ summarises target occupancy at decision time but loses information about *how* peers arrived there. The factorization is therefore a projection onto observable per-agent traces under that summary. Two specific failure modes arise: (i) inter-agent communication side-channels not logged through $\mathcal{C}_K$; (ii) coordinated peer policies whose decision at the C stage depends on hidden peer state not summarised by $\mathcal{O}_{-i}$.

4.3 Bottleneck shift

Empirical Claim 1 (bottleneck shift $M \leftrightarrow C$, slope-level). Assume scarcity ($N > T$ targets, see §2.1) and a peer-broadcast architecture with cadence $K \in \mathbb{N}$. Two stylized assumptions are posited: **(F)** memory freshness $\nu(\mathcal{M}_{-i}^{(K)})$ is non-decreasing pointwise as K decreases, and the probability

that agent i 's planner locks onto a reachable target is non-decreasing in ν ; **(O)** the probability that an arbitrary committed target is already in $\mathcal{O}_{-i}$ is non-increasing as K increases. Write $P(M^*|R^*; K)$ and $P(C^*|M^*; K)$ for the population-level conditional probabilities (averaged over agents and the seed distribution of the benchmark); $\hat{P}(\cdot)$ denotes the corresponding empirical-frequency estimator computed from logs. Under (F) and (O),

$$\frac{\partial P(M^*|R^*; K)}{\partial K^{-1}} > 0, \quad \frac{\partial P(C^*|M^*; K)}{\partial K^{-1}} < 0.$$

Both slope conditions are direct empirical predictions that are testable on logs through $\hat{P}$.

What this claim is and is not. Empirical Claim 1 is a *slope-level statement on conditional rates*: each of the two slopes is testable and is confirmed on the 35,640-run sweep (§4.5) and independently on a 100-run standard-MiniGrid portability sweep (§5). We report the slopes as *effect sizes with cluster-robust confidence intervals* (resampled over the 81 independent configuration cells rather than the correlated run count) in Appendix F.3. The claim's boundary with respect to the conditional product Φ is stated once, in the Scope-and-non-claims box (§1) and in §4.5.

Chronology. Empirical Claim 1 is post-hoc. It was formulated *after* an initial 12,960-run sweep conducted under four stage-agnostic hypotheses (Appendix F); the two slope conditions were then confirmed on the full 35,640-run sweep, on the $N = 12$ scale-up (Appendix E), and on the MiniGrid portability wrapper (§5). We state this explicitly: the verified slope signs are predictions on enlarged data, not a pre-registered prediction. A fully formal version is left for future work.

4.4 Sweep protocol

The four architectures are evaluated on a custom multi-agent grid (12×10 cells); each agent must reach a cell tagged `water_source`, choosing actions via a shared memory-driven planner so only the memory layer differs. The sweep crosses $N \in \{3, 5, 8\}$, $T \leq N$, three layouts (symmetric/asymmetric/random), three hazard densities, four architectures, eight peer cadences $K \in \{1, 2, 4, 8, 16, 32, 48, 64\}$, 40 seeds — 35,640 runs in ≈ 22 min on 8 cores. Per-agent per-tick Q/R/M/C indicators are logged (Appendix C); episode-level starred events are the disjunction over ticks. A separate oracle-intervention block on scarcity cases $(N, T) \in \{(3, 2), (5, 3), (8, 5)\}$ at five cadences evaluates oracle R (memory returns ground-truth water positions), oracle M (planner locks onto nearest real water), and oracle C (completion of a real materialised lock is guaranteed; Appendix F.12).

4.5 Results: stage decomposition and bottleneck shift

The 35,640-run sweep is reported through four observations, structured to mirror Empirical Claim 1: (i) the conditional decomposition isolates the M and C stages as the differentiating links; (ii) the bottleneck shift appears directly as a negative within-cadence correlation between $P(M^*|R^*)$ and $P(C^*|M^*)$, peaking at $K = 4$ and decaying to noise at $K = 64$; (iii) on mean time-to-success, the resulting trade-off produces an interior optimum at $K = 8$ — robust in location but shallow in margin, with the advantage over the adjacent fast cadences not resolved at this sample size (Appendix F.3); (iv) on the conditional product Φ , the same trade-off produces a monotone trend rather than an interior peak — a summary statistic on which we make no formal claim. A learned-communication baseline and a multi-agent oracle-intervention loop close the subsection.

4.5.1 Conditional decomposition reveals where each architecture fails.

Figure 4 shows per-architecture conditional Q/R/M/C rates; full numerics with bootstrap CIs are in Appendix F.1, Table 7. Retrieval conditional $P(R^*|Q^*)$ is saturated; the differentiating signal lives at M and C. Shared-bus, central-aggregator, and fast-broadcast peer ($K \leq 4$) have high $P(M^*|R^*)$ but low $P(C^*|M^*)$; independent and slow-broadcast peer ($K \geq 16$) show the reverse — the $M \leftrightarrow C$ trade-off predicted by Empirical Claim 1.

4.5.2 Bottleneck shift and interior efficiency optimum.

Across all peer runs, Spearman of $P(M^*|R^*)$ vs K is -0.53 (95% cluster-robust CI $[-0.59, -0.47]$) and of $P(C^*|M^*)$ vs K is $+0.43$ (CI $[+0.37, +0.50]$); both are moderate effect sizes whose

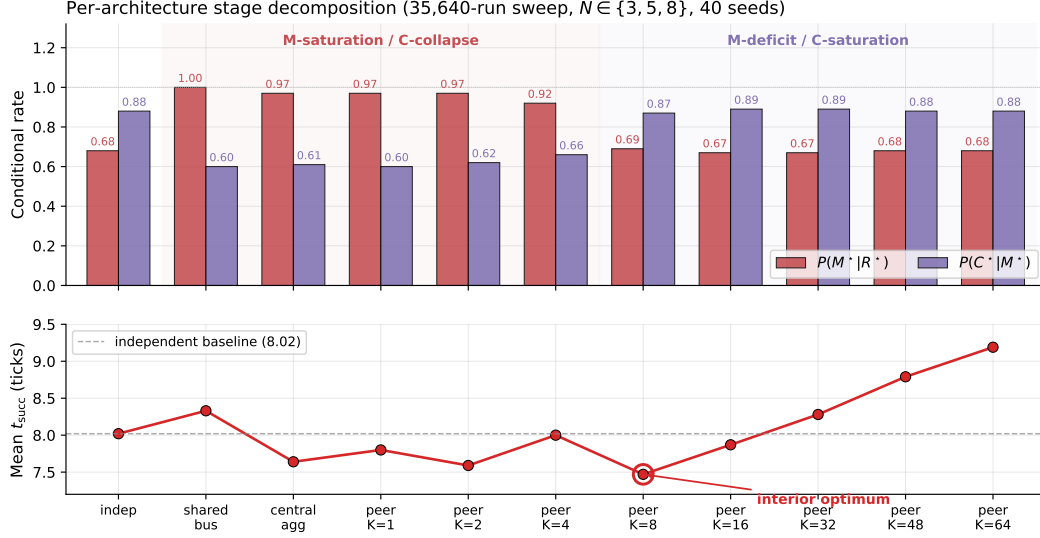


Figure 4: Per-architecture conditional rates (top) and mean time-to-success (bottom) from the 35,640-run sweep. Shared bus, central aggregator, and fast peer broadcast ($K \leq 4$) live in the M-saturation / C-collapse regime; independent and slow peer ($K \geq 16$) live in the M-deficit / C-saturation regime. Mean time-to-success has an interior minimum at $K=8$ (7.47 ticks), below the independent-agent baseline (8.02); the location is robust but shallow — the margin over the adjacent fast cadences is not resolved (Appendix F.3).

401 CIs exclude zero under resampling of the 81 configuration cells (Appendix F.3), match-
 402 ing the slope conditions of Empirical Claim 1. We report these rather than p -values,
 403 which are uninformative at this correlated sample size. The stronger empirical signature
 404 is at the per-configuration level: Pearson of $P(M^*|R^*)$ vs $P(C^*|M^*)$ within a fixed K is
 405 $-0.43, -0.54, -0.61, -0.19, -0.05, -0.04, -0.02, -0.02$ at $K=1, 2, 4, 8, 16, 32, 48, 64$, peaking
 406 at $K=4$ and decaying to noise by $K=64$ (Figure 8, Appendix F.1). Configurations where M is
 407 high systematically have low C : the bottleneck migrates between the two stages within the same
 408 (N, T , layout, hazard, seed) cell as K varies. On the efficiency metric, mean t_{succ} has a U-shape with
 409 interior minimum at $K=8$ (7.47 ticks vs 8.02 for the independent baseline); the minimum’s location
 410 is robust but shallow — its margin over the adjacent fast cadences ($K=1, 4$) is not resolved under
 411 cell-level resampling (paired-bootstrap CIs in Appendices F.1 and F.3). Scaling to $N=12$ sharpens
 412 the signature: the peak Pearson rises to -0.70 and migrates outward to $K=8$, consistent with
 413 stronger occupancy coupling at higher N delaying the cost-regime transition (Appendix E, Figure 6).

414 **Provenance stratification: the mechanism behind the shift.** Annotating each M^* lock with the
 415 *provenance* of the evidence behind it makes the bottleneck shift mechanistic rather than correlational.
 416 Let φ be the foreign share of the evidence mass supporting the locked candidate, computed from
 417 the same trust/support weights the merge itself used and frozen at lock time, and call a lock a *warp*
 418 (W^*) when $\varphi \geq 0.8$: the agent is committing to a target it knows essentially only from peers. On
 419 a dedicated peer sweep over the scarcity cells (random layouts, 20 seeds, $K \in \{1, 2, 4, 8, 16\}$) the
 420 warp share $P(W^*|M^*)$ falls from 0.55 at $K=1$ to 0.11 at $K=8$ while the own-evidence stratum
 421 completes at a flat 0.26–0.34 and the warp stratum completes below 0.05 throughout: *the C-collapse*
 422 *of fast sharing is concentrated in the warp stratum*. What cadence controls, mechanically, is how
 423 much of the collective’s materialisation runs on foreign evidence — and that is precisely the stratum
 424 that loses the occupancy race. The same stratification reproduces on the continuous VMAS bridge
 425 ($0.72 \rightarrow 0.00$ over $K=1/4/16$) and on an LLM text substrate ($0.42 \rightarrow 0.08$ over $K=2/8$), with the
 426 curves nearly coinciding across substrates. A full provenance-conditioned analysis (counterfactual
 427 attribution by masked replay, a closed-form staleness horizon with a registered 6/6-exact hold-out,
 428 and a decentralized repair protocol) appears in a companion manuscript; here the stratification serves
 429 as the mechanistic reading of Empirical Claim 1.

430 **What we do not claim about Φ .** The conditional product $\Phi = P(M^*|R^*) \cdot P(C^*|M^*)$ does
 431 *not* have an interior peak in the tested range (mean-of-products estimator increases monotonically
 432 from 0.569 at $K=4$ to 0.603 at $K=64$, converging to the independent-agent boundary 0.605). The
 433 slope-level statement of Empirical Claim 1 is on the conditional rates themselves, not on Φ ; we
 434 make no formal claim about the shape of $\Phi(K)$. Jensen-gap detail and POM-MOP comparison in
 435 Appendix F.1.

436 **Learned-communication baseline: stage events are structurally degenerate even at competi-**
 437 **itive success.** A CommNet-style policy (Sukhbaatar et al., 2016) trained with PPO (Schulman
 438 et al., 2017) on the same scenario ($N=3$, $T=2$) reaches 0.667 held-out success ($n=100$ seeds)
 439 — matching the symbolic peer architecture at $K=8$ (0.65). Despite this, the Q/R/M/C profile re-
 440 mains structurally degenerate: $Q^*=1.00$ saturated by construction; $R^*=0.997$, $M^*=0.997$ collapsed
 441 together ($P(M^*|R^*)=1.00$). This is the verified structural claim: a real-valued comm vector is
 442 never informatively empty (Q saturates) and lacks an explicit target-lock interface (R/M merge),
 443 regardless of training budget. The multi-agent counterpart of the single-agent learned-BC analysis
 444 (§3.3). Full architecture, PPO hyperparameters, and side-by-side REINFORCE vs PPO comparison
 445 in Appendix H.

446 **LLM model sweep: the logger localizes a real failure in language agents.** Table 3 runs the
 447 same Q/R/M/C event contract on two local LLM agents (qwen3:4b and llama3.1 via Ollama) on
 448 TextNav, sweeping the step budget over $\{6, 12, 25\}$ (8 episodes each; the task needs ~ 10 low-level
 449 steps). At a generous budget (12, 25) both agents succeed and all four stages fire. At a budget
 450 below the requirement (6), success drops to 0/8 for *both* models — and the decomposition shows
 451 *where*: $Q^*=R^*=M^*=1.00$ (the agents form the query, retrieve, and lock the correct target every
 452 episode) while $C^*=0.00$ (none complete in time). The framework localizes the failure precisely to
 453 a budget-limited completion (C) rather than to query formation, retrieval, or materialization; that
 454 both models fail identically shows the failure is structural, not a model artifact. This is the diagnostic
 455 property the framework claims, now exhibited on *real language agents*: an otherwise opaque success
 456 collapse ($1.00 \rightarrow 0.00$) is resolved into a specific failing stage. (Instrumentation note: a naive
 457 qwen3:4b/Ollama chat-template call initially returned empty response strings because the model
 458 spent the short structured-output budget in a separate thinking field; we use raw structured completion
 459 for qwen3:4b and a first-line parser, recorded for reproducibility.)

460 **The multi-agent phenomenology reproduces on a live LLM collective.** Beyond the single-agent
 461 budget study, the same event contract instruments a collective of $N=3$ LLM agents (llama3.1)
 462 sharing symbolic place memory by peer broadcast under scarcity ($M=2$ takeable targets, four layouts,
 463 $K \in \{2, 8\}$), with all predictions registered before the runs. The cadence phenomenology of §4.5
 464 carries over to the language substrate: a paired comparison against the no-communication baseline on
 465 identical layouts shows raw sharing *lowering* success from 2/3 to 1/3 per episode — the occupancy-
 466 racing C-collapse, now produced by agents whose queries, tag extraction, and commitments are
 467 LLM-made — and target collisions appear in 7/8 shared-memory episodes. Attaching the reservation-
 468 and-rollback repair protocol (the constructive answer to open problem 3 of Appendix F.8) recovers
 469 part of the loss ($0.333 \rightarrow 0.417$ at both cadences, rollback latency again $\approx K$). Two interface findings
 470 transfer to any LLM-over-symbolic-memory design: LLM query formers free-generate retrieval
 471 tags outside the memory’s vocabulary (requiring tolerant retrieval gates), and small local models are
 472 reliable at semantic commitment but not at coordinate arithmetic or systematic exploration (requiring
 473 layered execution). Full protocol and provenance analysis in the companion manuscript.

474 Scope: these are controlled studies on local models; scaling the same contract to full
 475 Voyager/MemGPT-style agents on richer tasks with multiple failure modes is developed next (external-
 476 framework validation) and in Appendix F.10.

477 **Multi-agent oracle interventions close the diagnose→intervene→verify loop.** On the scarcity
 478 cases, mean t_{succ} at small/intermediate cadence ($K \in \{1, 2, 4\}$) drops from ~ 10 –11 in baseline to
 479 ~ 7.2 under either oracle R or oracle M, and further to ~ 6.1 under oracle C, which additionally
 480 removes post-lock travel; at slow cadence ($K=16$) the baseline is already within ~ 1.2 of the R/M
 481 oracle floor (Table 14 in Appendix F.11). Crucially, oracle C gives the largest time reduction but
 482 the smallest success-rate gain (Appendix F.12): in scarcity, completion costs time while retrieval

Table 3: LLM TextNav step-budget sweep: Q/R/M/C localizes a completion-limited failure in two local Ollama agents. Eight episodes per (model, step limit), temperature 0.0; the task needs ~ 10 low-level steps. At a budget below that requirement (step limit 6) success drops to 0.00 for both models, yet $Q^*=R^*=M^*=1.00$ (the agents form the query, retrieve, and lock the correct target every episode) while $C^*=0.00$: the framework localizes the failure to the C stage, not to Q/R/M. Both models behave identically, so the failure is structural (budget), not model-specific. Artifacts: tmp/llm_steplimit_{6,12,25}.

Model	Step limit	Success	Q^*	R^*	M^*	C^*
qwen3:4b	6	0.00	1.00	1.00	1.00	0.00
llama3.1	6	0.00	1.00	1.00	1.00	0.00
qwen3:4b	12	1.00	1.00	1.00	1.00	1.00
llama3.1	12	1.00	1.00	1.00	1.00	1.00
qwen3:4b	25	1.00	1.00	1.00	1.00	1.00
llama3.1	25	1.00	1.00	1.00	1.00	1.00

and materialisation decide success. The cadence-induced excess time at small/intermediate K is attributable to the $M \leftrightarrow C$ transition; at large K the bottleneck has moved into solo exploration.

The multi-agent block thus delivers its half of the argument. The cadence K is, mechanically, the rate at which private memories consolidate into a collective view; the data show that this rate has a non-trivial interior optimum ($K=8$ on t_{succ}), that its cost channel is occupancy coupling, and that both effects are visible stage-by-stage on the same Q/R/M/C chain used for single agents. Consolidation is not only the single-agent limit; in a collective it is a tunable, measurable design axis.

5 Limitations and future work

Portability across substrates: slope conditions hold on MiniGrid. The slope conditions of Empirical Claim 1 also hold on a standard-MiniGrid substrate. A FourRooms-based multi-agent wrapper (built from `minigrid.core.grid.Grid + Wall/Floor/Lava` primitives, identical operational Q^*, R^*, M^*, C^* definitions; `experiments/minigrid_multiagent_wrapper/`) was used to run a 100-run portability sweep ($K \in \{1, 4, 8, 16, 64\}$, 20 seeds, $N=3, T=2$): Spearman $P(M^*|R^*)$ vs K is -0.50 and Spearman $P(C^*|M^*)$ vs K is $+0.50$ (moderate rank correlations; this is a small 100-run sweep, so we report the effect sizes and do not lean on significance). The interior t_{succ} minimum does not separate at this minimal sweep size (5 cadences, 20 seeds, one hazard density); a denser sweep is the natural next step. The two-substrate seam is therefore closed at the slope-level claim.

Empirical Claim 1 is slope-level on conditional rates only. The two slope conditions hold as moderate, cluster-robust effect sizes (Appendix F.3); we make no formal claim on the shape of the conditional product Φ (§4.3). A future statement directly in terms of the time-cost surface, or a derivation under a generative model (Poisson-renewal memory, birth-death occupancy), would convert the empirical claim into a theorem.

Estimability under hidden state. With unlogged in-episode query rewrites or peer side-channels, Definitions 1 and 2 are projections onto logged traces, not full identification.

Shared coordinate frame. All multi-agent results assume agents interpret places in a common coordinate frame: cross-agent place identity is coordinate proximity, with semantic profiles as a tie-breaker inside a spatial radius. The place-correspondence problem — whether two agents’ records denote the same place at all — is therefore bracketed here, not solved. A companion manuscript removes the assumption constructively (fingerprint-based place identity with consensus frame recovery up to translation and 90° rotation) and shows the Q/R/M/C measurements, including the cadence slopes and the staleness horizon, unchanged on top of the recovered frames; the harder variants (continuous $SE(2)$, appearance noise, symbol alignment) remain open.

Learned baselines. The CommNet PPO baseline reaches 0.667 success on the scarcity scenario — competitive with symbolic peer at $K=8$ (0.65) — and confirms Q-saturation ($Q^*=1.00$) and R/M collapse ($P(M^*|R^*)=1.00$) at competitive success. The structural prediction now holds verified, not asserted; ablating on architectures with explicit symbolic target-lock channels is left for future work.

Scale. Multi-agent sweep bounded at $N=12$ (Appendix E; signature sharpens, peak Pearson -0.70 at $K=8$, interior t_{succ} minimum 4.77). Denser scaling at $N \geq 16$ is future work. **Continuous substrate.** A preliminary VMAS (Bettini et al., 2022) check reproduces *both* slope signs of Empirical Claim 1 in the coupling regime (Spearman -1.00 and $+1.00$ over $K \leq 16$, 40 seeds; Appendix F.7), reusing the symbolic memory through a discretisation bridge. This is a directional signal only — a perfect rank correlation over five monotone points, one configuration, discretised memory — so it indicates the shift is not an artefact of grid cells, but a full continuous-state validation with cluster-robust CIs remains future work.

6 Broader impacts

The paper proposes diagnostic evaluation for memory-based agents in simulated navigation. Its positive impact is methodological: making memory-based failures more measurable across architectures, and making the cadence axis of distributed multi-agent memory tunable on an empirically grounded criterion. Potential negative impacts are indirect and would arise if similar diagnostics were used to improve agents in safety-critical settings (drone swarms, autonomous vehicles, distributed sensor networks) without adequate safeguards. The current work is benchmark-level infrastructure, not a deployment-ready system.

7 Conclusion

We proposed Q/R/M/C, a four-stage decomposition of memory-based navigation whose factors are estimable from standard trajectory logs, and used it as a diagnostic instrument across symbolic, learned, LLM-driven, single-agent, and multi-agent traces. The main result is not a new agent architecture: it is a reproducible logging and estimation protocol that separates query formation, retrieval satisfaction, target materialization, and post-retrieval completion. Across the experiments, this protocol localizes failures that success-only evaluation cannot, exposes degenerate observability in implicit learned policies, and makes memory-sharing cadence measurable as a stage-level design variable.

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627 A Formal proofs

628 A.1 Argument for Definition 1 (single-agent estimability)

629 For each episode $i \in \{1, \dots, n\}$ let $Q_i^*, R_i^*, M_i^*, C_i^*$ denote the nested episode-level indicators. By
630 construction these are measurable functions of the runtime trajectory log and satisfy $C_i^* \Rightarrow M_i^* \Rightarrow$
631 $R_i^* \Rightarrow Q_i^*$. The empirical estimators

$$\hat{p}_Q = \frac{1}{n} \sum_i Q_i^*, \quad \hat{p}_{R|Q} = \frac{\sum_i R_i^*}{\sum_i Q_i^*}, \quad \hat{p}_{M|R} = \frac{\sum_i M_i^*}{\sum_i R_i^*}, \quad \hat{p}_{C|M} = \frac{\sum_i C_i^*}{\sum_i M_i^*}$$

632 are consistent by the chain rule plus the conditional stage-Markov property, and the empirical product
633 converges to semantic-path completion. $\square$

634 A.2 Argument for Definition 2 (multi-agent estimability)

635 Conditioning sets are enlarged by $\mathcal{M}_{-i}^{(K)}$ and $\mathcal{O}_{-i}$. Under the enlarged stage-Markov property, the
636 proof of Definition 1 transfers verbatim and each enlarged conditional is a Bernoulli probability on
637 a measurable event. The qualifications stated in §4.2 (“When Definition 2 may fail”) describe two
638 specific situations in which the enlarged stage-Markov property is violated. $\square$

639 A.3 Argument for Empirical Claim 1 (slope-level statement)

640 Empirical Claim 1 is an empirical claim, not a theorem. The argument below is the qualitative
641 justification that motivates the two slope signs; the data are what verify them. We deliberately do not
642 call this a proof.

643 *Slope (a), under (F).* As K decreases, $\nu(\mathcal{M}_{-i}^{(K)})$ is non-decreasing pointwise. Conditional on
644 a successful retrieval, the probability of locking onto a reachable target is non-decreasing in ν .
645 Therefore $\partial P(M^*|R^*; K)/\partial K^{-1} > 0$.

646 *Slope (b), under (O).* As K decreases, peers receive updated memory simultaneously and con-
647 verge on the same closest target; the marginal occupancy $\Pr[\text{target} \in \mathcal{O}_{-i}]$ rises. Therefore
648 $\partial P(C^*|M^*; K)/\partial K^{-1} < 0$.

649 *Note on the conditional product Φ .* Empirical Claim 1 makes no statement about the shape of
650 $\Phi(K) = P(M^*|R^*; K) \cdot P(C^*|M^*; K)$. Under (F) and (O) the slope of Φ in K^{-1} is the sum of two
651 opposing terms, and whether it changes sign depends on quantitative magnitudes that are not pinned
652 down by (F) and (O) alone. In our data Φ is monotone in K over $\{4, 8, 16, 32, 48, 64\}$ and converges
653 to the independent-agent boundary $\Phi_\infty = 0.605$. The practically relevant efficiency surface — mean
654 time-to-success — does have an interior minimum at $K = 8$; see §4.5. $\square$

655 A.4 The query–retrieval Galois connection: statement, argument, scope

656 **Fact (standard).** Let G and Σ be sets and $I \subseteq G \times \Sigma$ a relation. For $B \subseteq \Sigma$ define $\text{ext}(B) =$
657 $\{p \in G : \forall \tau \in B, (p, \tau) \in I\}$, and for $A \subseteq G$ define $\text{int}(A) = \{\tau \in \Sigma : \forall p \in A, (p, \tau) \in I\}$.
658 Then for all $A \subseteq G, B \subseteq \Sigma$:

$$A \subseteq \text{ext}(B) \iff B \subseteq \text{int}(A);$$

659 both maps are antitone, both composites $\text{ext} \circ \text{int}$ and $\text{int} \circ \text{ext}$ are closure operators, and the closed
660 pairs (A, B) with $A = \text{ext}(B)$ and $B = \text{int}(A)$, ordered by extent inclusion, form a complete lattice
661 — the concept lattice of the context (G, Σ, I) (Ganter and Wille, 1999).

662 **Argument.** $A \subseteq \text{ext}(B)$ unfolds to $\forall p \in A \forall \tau \in B : (p, \tau) \in I$, a condition symmetric in A
663 and B , which is exactly $B \subseteq \text{int}(A)$. Antitonicity and the closure properties follow by the standard
664 one-line arguments (Ganter and Wille, 1999, Ch. 1). Viewing the power sets as thin categories (one
665 with reversed order), the displayed equivalence says precisely that int and ext form an adjoint pair; a
666 Galois connection is an adjunction between posets (Mac Lane, 1998, Ch. IV). $\square$

667 **Instantiation and scope.** In our memory, take G to be the place records, Σ the tag alphabet,
668 and $I_\theta = \{(p, \tau) : \text{the per-place tag table assigns } \tau \text{ to } p \text{ with confidence } \geq \theta\}$ (Appendix D); the
669 intent of a query is $B = q^{\text{req}}$ and R-availability is $\text{ext}(B) \neq \emptyset$. Three deliberate scope restrictions.
670 (i) *Graded scores.* The deployed retrieval score is graded (the weighted log-score over required,
671 preferred, and penalty tags of Appendix D), so the crisp connection above formalizes the thresholded
672 required-tag skeleton of candidacy; the graded analogue is the theory of fuzzy Galois connections
673 (Bělohávek, 1999) and we do not use it here. (ii) *Non-monotone updates.* Dirichlet–multinomial
674 updates are not literally monotone in I_θ : contrary evidence can remove an incidence. On the
675 reported benchmarks the binding failure is absence of evidence rather than contradiction (91% empty
676 extents, 96.9% precision when non-empty, §3.5), so the monotone-growth reading is the empirically
677 relevant regime, and we flag the exception rather than assume it away. (iii) *No structure for M or C .*
678 Materialization is a satisficing choice function on non-empty extents and Completion is a control
679 problem under occupancy; neither carries the order structure, and we attach no categorical claims to
680 them. The reading earns its place by naming the failure mode exactly (a queried intent whose extent
681 is empty lies outside the realized concept lattice), by giving consolidation a precise object (growth of
682 the context, hence of every extent), and by locating the cadence axis structurally: memory coupling
683 $\mathcal{M}_{-i}^{(K)}$ merges peer contexts at rate K^{-1} , while occupancy $\mathcal{O}_{-i}$ is extra-contextual — which is the
684 structural content behind the two slope signs of Empirical Claim 1.

685 B An options / hierarchical-MDP reading of the stages

686 This appendix develops the dynamic reading pointed to from §2.3. It introduces no new algorithm: it
687 names the object the pipeline already computes and makes the Φ -versus- t_{succ} split a property of that
688 object. Where the order-theoretic reading (Appendix A.4) fixes the *static* content of the stages, this
689 one addresses time and cost.

690 **Q/R/M/C as an audit of an option.** Each query instantiates one temporally extended action—an
691 option $o = \langle \mathcal{I}, \pi, \beta \rangle$ in the sense of Sutton et al. (1999): an initiation set $\mathcal{I}$, an internal policy π , and
692 a termination condition β . Standard trajectory-level analysis treats an option as a black box scored by
693 its return. The Q/R/M/C decomposition is instead an audit of the option’s *internal* structure, and it
694 lines up edge-for-edge with the factorization $Q^* \rightarrow R^* \rightarrow M^* \rightarrow C^*$:

695 $Q^* \rightarrow R^*$ (**initiation**). The option “reach a place satisfying q ” is well-initiated only when its initiation
696 set is non-empty. In the vocabulary of §2.3 this is exactly $\text{ext}(q^{\text{req}}) \neq \emptyset$: an empty extent is not
697 a bad choice of option but the absence of any admissible one, so R fails before a target is ever
698 locked.

699 $R^* \rightarrow M^*$ (**policy over options**). Materialization is the high-level policy that commits to a single
700 option out of the admissible set—the target lock $g^* = \chi(\text{ext}(q^{\text{req}}))$ under the satisficing choice
701 function χ of §2.3.

702 $M^* \rightarrow C^*$ (**execution to termination**). Completion is the roll-out of the internal policy π until the
703 termination condition β fires (arrival at g^* , or timeout). This is the only edge whose failure modes
704 are extra-semantic: controller competence and, in a collective, occupancy $\mathcal{O}_{-i}$.

705 So Q/R/M/C is, on this reading, a stage-resolved instrument for the internal structure of a hierarchical-
706 MDP option generator, rather than a bespoke logger for a single environment.

707 **Why the optimum lives on t_{succ} and not on Φ : a semi-MDP reading.** In the options / semi-MDP
708 formalism, an option is evaluated by a return accumulated over a *random* execution duration τ ,

$$R(s, o) = \mathbb{E} \left[\sum_{t=0}^{\tau-1} \gamma^t r_{t+1} \mid s_0 = s, o \right],$$

709 where τ is the number of low-level steps until β fires (Sutton et al., 1999). The chain probability
710 $\Phi = P(M^*|R^*) \cdot P(C^*|M^*)$ records only *whether* the option-audit succeeds; it is by construction
711 blind to τ , which is the cost of traversing it. The cost-bearing metric t_{succ} is τ . This is the structural
712 reason—and not merely an empirical coincidence—that the resource-rational reading of §2.3 (Lieder
713 and Griffiths, 2020) locates an interior optimum on t_{succ} ($K=8$) while Φ stays monotone: the two
714 metrics measure different faces of the same option, success versus duration.

715 **The bottleneck shift as multi-agent option collision.** The dynamic reading makes the $M \leftrightarrow C$
716 trade-off of Empirical Claim 1 physically concrete. Faster broadcast (smaller K) merges peer contexts
717 $\mathcal{M}_{-i}^{(K)}$ more often, which can only enlarge each agent’s extent and therefore raises the probability
718 that a target is admissible and locked: $P(M^*|R^*)$ increases. But because agents share a spatial basis
719 and a tag alphabet (§2.1), fresher shared memory also *synchronizes* the high-level policy: several
720 agents commit to the same option over the same scarce target. Under occupancy $\mathcal{O}_{-i}$ this inflates
721 the execution duration τ and depresses $P(C^*|M^*)$. Cadence K thus regulates the dispersion of the
722 agents’ option distribution: rare exchange desynchronizes option selection and shortens τ , giving the
723 minimum of t_{succ} at an interior K . The bottleneck is therefore grounded in a competition between
724 initiation-set growth (semantic, contextual) and option-collision cost (extra-semantic, occupancy),
725 exactly the two edges the cadence protocol $\mathcal{C}_K$ acts on in Figure 2.

726 **Belief–desire–intention gloss and scope.** The high-level state audited above can be named in
727 belief–desire–intention terms, which is the form used by the collective-memory companion: *belief* is
728 the realized concept lattice of §2.3 (what is retrievable), *desire* is the state-conditioned predicate that
729 selects the query q (e.g. a thirst variable switching the required tags to `water_source`), and *intention*
730 is the locked option g^* . The hierarchical state is then $h = (\text{lattice}, \text{desire}, g^*, s)$ with s the low-level
731 physical state, and the three failure loci separate as before: an empty extent is a belief/ R failure, a
732 non-empty extent with no stable lock is an M failure, and a lock unreachable under occupancy is a C
733 failure. Two scope caveats. We claim no options-*learning* result: the pipeline is a fixed satisficing
734 option generator (Simon, 1955) and Q/R/M/C audits it. And we attach the option structure only
735 to the semantic path— $\mathcal{I}$ to $Q \rightarrow R$, (π, β) to $M \rightarrow C$ —while the semi-MDP return is used as an
736 explanatory lens for the Φ -versus- t_{succ} split, not fitted to the data. Neither this options reading nor the
737 order-theoretic reading of Appendix A.4 is claimed as a contribution: both are standard constructions
738 used as interpretive lenses on the measurement protocol, which is the paper’s contribution. Their
739 value is that each makes a testable statement on the logs—an empty extent localises the R failure,
740 and the semi-MDP duration τ predicts that the optimum appears on t_{succ} and not on Φ .

741 C Operational vs. nested stage estimators

742 **Operational diagnostic rates.** The main benchmark logs per-query and per-episode diagnos-
743 tic quantities directly from runtime traces (query formed, retrieval satisfied, target materialized,
744 completion after retrieval). These are operational rates, not direct multiplicative factors.

745 **Nested stage estimators for factorization.** For the factorization, episode-level nested events
746 from the first failure stage recorded in the runtime taxonomy are used: Q^*, R^*, M^*, C^* , nested by
747 construction. The consistency audit in §3.2 reports $\hat{P}(Q^*)$, $\hat{P}(R^*|Q^*)$, $\hat{P}(M^*|R^*)$, $\hat{P}(C^*|M^*)$,
748 their product, and its gap to semantic-path completion.

749 C.1 Multi-agent operational definitions

750 For the multi-agent sweep, every metric is defined per agent and per tick, then aggregated. Let
751 $\text{Targets}_i(t)$ be the candidate set returned by the memory query of agent i at tick t , $\text{Locked}_i(t)$ be the
752 agent’s planner-internal locked target, and $\mathcal{W}$ the ground-truth target set.

753 **Per-tick events.** $Q_i(t) = \mathbf{1}[\text{Targets}_i(t) \neq \emptyset]$; $R_i(t) = \mathbf{1}[\exists w \in \mathcal{W} : \min_c \|c - w\| \leq \epsilon]$ ($\epsilon = 0.6$);
 754 $M_i(t) = \mathbf{1}[\text{Locked}_i(t) \neq \emptyset \wedge \min_w \|\text{Locked}_i(t) - w\| \leq \epsilon]$.

755 **Episode-level starred events.** $Q_i^* = \bigvee_t Q_i(t)$; analogously for R^*, M^* ; $C_i^* =$
 756 $\mathbf{1}[\text{agent } i \text{ reached a target}]$.

757 **Aggregation.** Per-run rates are averaged over the N agents; per-configuration rates are averaged
 758 over the 40 seeds. Conditional rates $P(R^*|Q^*)$, $P(M^*|R^*)$, $P(C^*|M^*)$ are computed as ratios of
 759 episode-level counts within each configuration before averaging across configurations.

760 D Memory and task schema

761 D.1 Full notation table

762 Symbols are fixed throughout the paper.

763 **Stages.** Q, R, M, C — the four stages: **Q**uery formation, **R**etrieval, **M**aterialization, **C**ompletion
 764 after retrieval. They name both the stage and (in expressions like “the M link”) the corresponding
 765 edge of the chain.

766 **Events.** $Q^*, R^*, M^*, C^* \in \{0, 1\}$ — episode-level starred events recording whether each stage
 767 succeeded (Appendix C). Nested by construction: $Q^*=1 \Leftarrow R^*=1 \Leftarrow M^*=1 \Leftarrow C^*=1$.

768 **Probabilities.** $P(\cdot)$ — population-level probability (averaged over the seed distribution and, in the
 769 multi-agent case, over agents). $\hat{P}(\cdot)$ — empirical-frequency estimator computed from logs.

770 **Single-agent.** q — query; $\mathcal{M}$ — memory state; Y — end-to-end task success (can exceed C^*).

771 **Multi-agent.** N — number of agents in the population. T — number of targets in the environment
 772 (constraint $T \leq N$; “scarcity” means $N > T$). i — agent index. $\mathcal{M}_i$ — agent i ’s private memory. q_i
 773 — agent i ’s query.

774 **Coupling.** $\mathcal{C}_K$ — the peer-broadcast communication protocol with cadence $K \in \mathbb{N}$ (every K
 775 ticks each agent broadcasts a memory snapshot, §4.1). $\mathcal{M}_{-i}^{(K)}$ — the latest peer memory snapshots
 776 accessible to agent i under $\mathcal{C}_K$ (the *memory coupling* channel). $\mathcal{O}_{-i}$ — the set of targets already
 777 claimed by other agents by the time i acts (the *occupancy coupling* channel).

778 **Slope variables.** $\nu(\mathcal{M}_{-i}^{(K)})$ — a freshness functional on peer memory; non-decreasing pointwise as
 779 K decreases (assumption **(F)** of Empirical Claim 1). The occupancy slope is assumption **(O)**.

780 **Summary statistics.** $\Phi(K) := P(M^*|R^*; K) \cdot P(C^*|M^*; K)$ — the conditional product, the chain
 781 probability $P(M^* \wedge C^* | R^*)$. POM (product-of-means) and MOP (mean-of-products) are two
 782 estimators of Φ ; they differ by $\text{Cov}(P(M^*|R^*), P(C^*|M^*))$, which is the Jensen gap (§4.5). t_{succ}
 783 — mean time-to-success in environment ticks.

784 D.2 Memory stack and task patterns

785 The memory stack stores symbolic place records (per-place tag confidences, counts, visitation,
 786 freshness, geometry), transition records (success, risk, cost statistics with fast/slow estimates),
 787 episodic traces (intent + state at each visit), and concept/relation records. At each step, the tokenizer
 788 emits (z_t, τ_t, e_t) from o_t . Per-place tag tables use a Dirichlet-multinomial update; transitions use
 789 exponentially smoothed statistics; episodic records attach the active intent and agent state. A planner
 790 query is answered by retrieving candidate places whose semantic profiles match the current task intent
 791 and state, via a weighted log-score $s(p | q_t, s_t) = \sum_{\tau \in q^{\text{req}}} \log P(\tau|p) + \alpha \sum_{\tau \in q^{\text{pref}}} \log P(\tau|p) -$
 792 $\beta \sum_{\tau \in q^{\text{pen}}} \log P(\tau|p) + \gamma u_{\text{epi}}(p, s_t)$ over required, preferred, and penalty tags. The four task
 793 families are defined by semantic place patterns: water search (target {water}), safe rest ({rest} with
 794 state-conditioning), goal-region rejoin ({goal-region}), hazard recovery ({post-hazard-exit}).

795 D.3 Single-agent supporting tables and figures

Table 4: Best semantic result per task family from the final 10-seed benchmark, with 95% bootstrap CIs. Hazard-recovery is bottlenecked by retrieval; goal-region remains downstream-bottlenecked.

Task	Best method	Assist	Success (95% CI)	Steps	Query (op.)	Retrieval (op.)	Mat. (op.)	Post (op.)
Water	water-cp	on	0.95 [0.88, 1.00]	17.99	0.71	0.71	0.71	0.70
Rest	rest-s2	on	0.93 [0.85, 0.99]	23.35	0.61	0.61	0.61	0.73
Goal region	goal-n1	off	0.54 [0.52, 0.56]	49.34	0.00	0.00	0.00	0.00
Hazard recovery	hazard-s7	on	0.40 [0.35, 0.45]	10.18	0.11	0.11	0.11	0.16

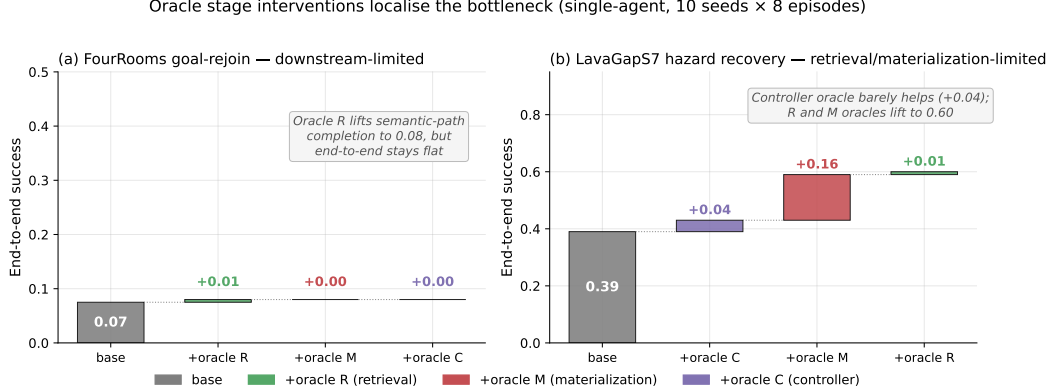


Figure 5: Oracle stage interventions on the two hardest single-agent task families (10 seeds $\times$ 8 episodes). **(a)** FourRooms goal-rejoin: replacing any single stage with an oracle leaves end-to-end success at 0.07; the bottleneck is downstream of the stages in the chain. **(b)** LavaGapS7 hazard recovery: oracle controller adds only +0.04, but oracle materialization adds +0.16 and oracle retrieval adds a further +0.01 to reach 0.60. The framework localises the bottleneck stage-by-stage.

796 E Portability and scale-up

797 E.1 BabyAI portability transfer (10 seeds, full benchmark)

798 A 10-seed portability benchmark on BabyAI-GoToObjMaze-v0 (Chevalier-Boisvert et al., 2018)
799 (8 episodes per seed, max-steps 200) measured all four single-agent methods used in the main
800 benchmark (random, greedy, babyai_semantic_node_v1, babyai_semantic_plan_v1) with
801 the same Q/R/M/C logger applied to BabyAI traces. The numbers in Table 5 confirm that (a) the
802 Q/R/M/C events emit non-trivially on the BabyAI substrate, (b) the two semantic variants expose
803 stage-separable structure where the non-semantic baselines have $Q^*=R^*=M^*=0$ (no semantic
804 queries are issued), and (c) the stage-rate ordering matches the MiniGrid benchmark (Table 1) within
805 bootstrap CIs. GoToObjMaze is a harder task substrate than the LavaGapS7/FourRooms benchmark
806 used in the main text — end-to-end success caps at 0.15 across all methods — which makes the stage
807 decomposition especially informative: the semantic stack is not winning on success but it is the only
808 method family producing measurable per-stage events.

Table 5: BabyAI portability benchmark: 10 seeds $\times$ 8 episodes on BabyAI-GoToObjMaze-v0, max-steps 200. The semantic-stack variants succeed at the same end-to-end rate as the greedy baseline (0.15) but are the only methods that produce non-zero Q/R/M/C events — the framework’s protocol transfers to the BabyAI substrate without modification. Q/R/M/C operational rates here are evaluated on the same denominator (number of decision points), so $Q=R=M$ for both semantic variants because GoToObjMaze emits one semantic target type per episode.

Method	Success	Steps	Q (op.)	R (op.)	M (op.)	Post (op.)
random	0.125 ± 0.06	182.7	0.000	0.000	0.000	0.000
greedy	0.150 ± 0.05	171.3	0.000	0.000	0.000	0.000
semantic node	0.150 ± 0.05	171.3	0.032	0.032	0.032	0.025
semantic plan	0.150 ± 0.05	171.3	0.032	0.032	0.032	0.025

Takeaway. The framework’s measurement protocol — not its task-specific results — transfers to a second substrate. End-to-end success on BabyAI is dominated by the much lower episode-level success ceiling of GoToObjMaze, not by the stage structure; conversely, the stage structure that the semantic stack exposes is identical in shape to what the same stack exposes on MiniGrid. This is exactly the substrate-independence property the methodology is intended to provide. Raw data: tmp/babyai_deeper_10seed/.

E.2 Multi-agent scale-up at $N = 12$

To check that the multi-agent results of §4.5 are not an artefact of the $N \leq 8$ range, an additional 8,640-run sweep was run at $N=12$, $T \in \{6, 8, 10\}$, the same three layouts, three hazard densities, eight peer cadences, and 40 seeds. Wall-clock ≈ 12 minutes on 8 cores. The conditional-rate signature of §4.5 persists and intensifies (Table 6).

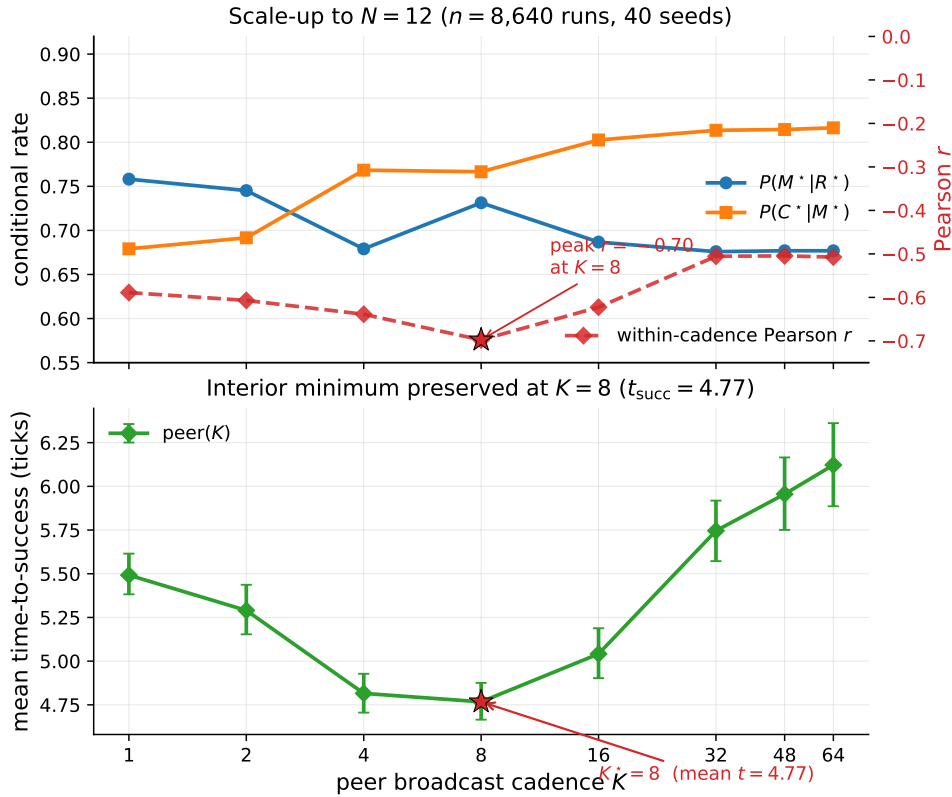


Figure 6: $N = 12$ scale-up bottleneck signature ($n = 8,640$ runs, 40 seeds). Top: the per-cadence mean rates move in the predicted directions in K ; the within-cadence Pearson correlation (red dashed, right axis) peaks at -0.70 at $K = 8$ (this within-cadence peak is what sharpens at $N=12$; the M-side K -slope itself is not cluster-robust at this agent count, Table 6). Bottom: mean time-to-success retains an interior minimum at $K = 8$ (4.77 ticks); as on the main sweep, the location is robust but the margin is shallow — the gap to the adjacent $K = 4$ (4.82 ticks) is ≈ 0.05 tick and is not resolved at this sample size. The bottleneck-shift peak migrates outward from $K = 4$ at $N \leq 8$ to $K = 8$ at $N = 12$, consistent with stronger occupancy coupling at higher agent count delaying the cost-regime transition.

F Extended diagnostics

F.1 Conditional-product detail and the Jensen gap

The conditional product $\Phi = P(M^* | R^*) \cdot P(C^* | M^*)$ in the 35,640-run sweep is monotonically increasing in K over $K \in \{4, 8, 16, 32, 48, 64\}$ and converges to the independent-agent boundary

Table 6: $N=12$ scale-up sweep numerical detail (peer architecture; 27 configuration cells). Effect sizes with cluster-robust 95% CIs (block bootstrap over cells): Spearman of $P(C^*|M^*)$ with K is $+0.31$ (CI $[+0.21, +0.41]$, excludes zero), while Spearman of $P(M^*|R^*)$ with K is only -0.14 (CI $[-0.28, +0.02]$, *not* resolved from zero at $N=12$). The C-slope of Empirical Claim 1 holds; the M-slope is directionally consistent but not cluster-robust at this agent count. What sharpens at $N=12$ is the within-cadence correlation and the t_{succ} minimum (Figure 6), not the M-side K -slope.

K	$P(M^* R^*)$	$P(C^* M^*)$	MOP	mean t_{succ}	corr
1	0.758	0.679	0.475	5.49	-0.59
2	0.745	0.692	0.474	5.29	-0.61
4	0.679	0.768	0.482	4.82	-0.64
8	0.731	0.767	0.539	4.77	-0.70
16	0.687	0.803	0.538	5.04	-0.62
32	0.676	0.814	0.538	5.75	-0.51
48	0.677	0.814	0.539	5.96	-0.51
64	0.677	0.816	0.541	6.12	-0.51

Table 7: Per-architecture conditional Q/R/M/C rates from the 35,640-run sweep (40 seeds). $P(R^*|Q^*)$ is saturated. POM = product-of-means; MOP = mean-of-products (the proper per-configuration expected product). The two summaries disagree systematically at small/intermediate K because the within-cadence correlation between $P(M^*|R^*)$ and $P(C^*|M^*)$ is strongly negative there (last column) — the bottleneck-shift signature. **Bold:** MOP-best of the peer cadences. Mean t_{succ} (last column) is the practically relevant efficiency measure; its interior minimum at $K=8$ is the predicted-shape result, robust in location but shallow in margin (adjacent fast cadences not resolved; Appendix F.3).

Architecture	$P(R^* Q^*)$	$P(M^* R^*)$	$P(C^* M^*)$	POM	MOP (95% CI)	corr	mean t_{succ}
independent	0.99	0.68	0.88	0.598	0.605 [0.597, 0.614]	+0.00	8.02
shared bus	1.00	1.00	0.60	0.600	0.595 [0.586, 0.604]	-0.10	8.33
central aggregator	1.00	0.97	0.61	0.592	0.572 [0.564, 0.580]	-0.48	7.64
peer ($K=1$)	1.00	0.97	0.60	0.585	0.573 [0.564, 0.581]	-0.43	7.80
peer ($K=2$)	1.00	0.97	0.62	0.594	0.576 [0.568, 0.584]	-0.54	7.59
peer ($K=4$)	1.00	0.92	0.66	0.599	0.569 [0.561, 0.577]	-0.61	8.00
peer ($K=8$)	0.99	0.69	0.87	0.597	0.586 [0.578, 0.595]	-0.19	7.47
peer ($K=16$)	0.99	0.67	0.89	0.592	0.590 [0.581, 0.598]	-0.05	7.87
peer ($K=32$)	0.99	0.67	0.89	0.595	0.598 [0.589, 0.606]	-0.04	8.28
peer ($K=48$)	0.99	0.68	0.88	0.598	0.602 [0.593, 0.611]	-0.02	8.79
peer ($K=64$)	0.99	0.68	0.88	0.599	0.603 [0.594, 0.611]	-0.02	9.19

824 $\Phi(K \rightarrow \infty) = 0.605$. Paired bootstrap: $\Delta(K=32-K=16) = +0.008$ (CI $[+0.006, +0.010]$),
825 $\Delta(K=48-K=16) = +0.011$, $\Delta(K=64-K=16) = +0.012$; the independent-agent level sits
826 above every peer K except $K=64$ (where the gap’s CI includes zero). This is a statement about the
827 *level* of Φ at the tested cadences, not about the *shape* of $\Phi(K)$: consistent with the monotone trend,
828 we make no claim of an interior extremum (§4.3).

829 The negative within-cadence covariance documented in §4.5 creates the Jensen-inequality gap
830 between Φ and the marginal product-of-means: $E[X] \cdot E[Y] > E[X \cdot Y]$ when $\text{Cov}(X, Y) < 0$. The
831 mean-of-products is reported as the primary summary because it absorbs the covariance correctly.
832 Figure 7 shows the K-sweep view of both conditional rates and mean t_{succ} ; Figure 8 shows the
833 per-configuration scatter that the negative covariance summarises.

834 F.2 Design decomposition and run-count reconciliation

835 The 35,640 figure decomposes exactly as in Table 8. The base design is the nine admissible (N, T)
836 scarcity pairs (for $N=3$: $T \in \{2, 3\}$; $N=5$: $T \in \{2, 3, 5\}$; $N=8$: $T \in \{2, 3, 5, 8\}$) crossed with three
837 layouts, three hazard densities, and 40 seeds, giving 3,240 base configurations. The peer architecture
838 is run at all eight cadences ($3,240 \times 8 = 25,920$); the three cadence-free architectures (independent,
839 shared bus, central aggregator) contribute 3,240 each. The 25,920-run figure used for the peer
840 cluster-robust analysis (Appendix F.3) is exactly the peer block; the initial 12,960-run exploratory
841 sweep (§4.3, “Chronology”) and the extended-cadence and $N=12$ sweeps (Appendix E) are separate
842 run sets and are not pooled with the main sweep.

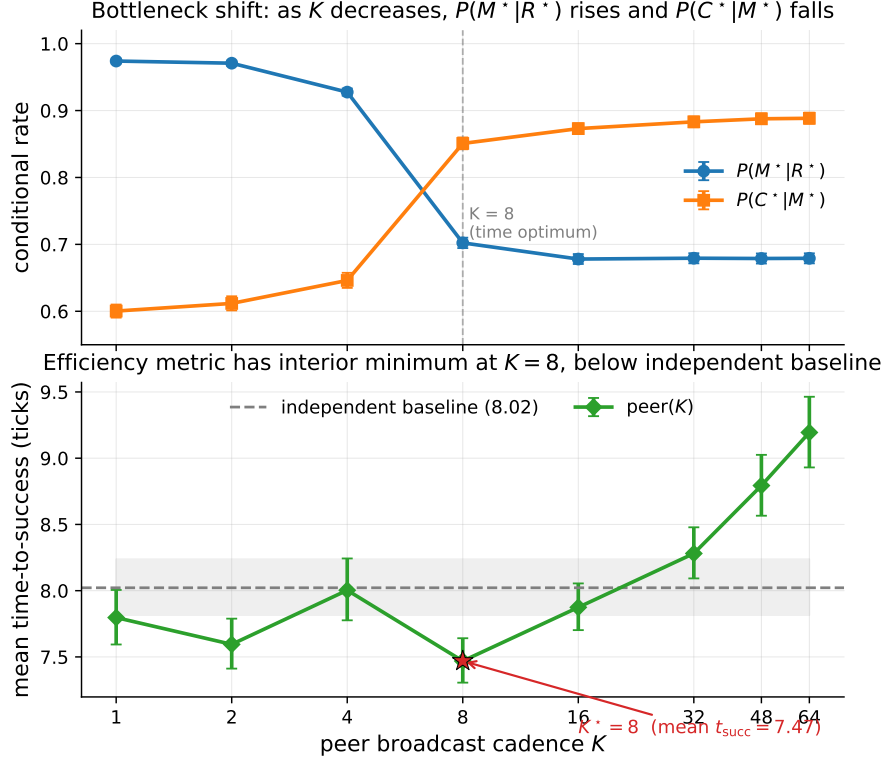


Figure 7: Bottleneck shift across eight peer cadences (40-seed sweep). Top: $P(M^*|R^*)$ falls and $P(C^*|M^*)$ rises in K (moderate, cluster-robust Spearman effect sizes -0.53 and $+0.43$; CIs in Appendix F.3). Bottom: mean time-to-success has an interior minimum at $K=8$ (7.47 ticks) below the independent-agent baseline (8.02, dashed); the location is robust but its margin over the adjacent fast cadences ($K=1, 4$) is not resolved (Appendix F.3). Error bars and shaded band are 95% bootstrap CIs.

Table 8: Run-count reconciliation for the main multi-agent sweep (35,640 runs). Counts are recomputed directly from the released run log.

Factor	Levels	Count
(N, T) scarcity pairs	$(3, 2)(3, 3)(5, 2)(5, 3)(5, 5)(8, 2)(8, 3)(8, 5)(8, 8)$	9
Layouts	symmetric, asymmetric, random	3
Hazard densities	0.0, 0.05, 0.10	3
Seeds		40
Base configurations	$9 \times 3 \times 3 \times 40$	3,240
Peer (all 8 cadences K)	$3,240 \times 8$	25,920
Independent / shared / central	$3 \times 3,240$	9,720
Total		35,640

843 F.3 Effect sizes and cluster-robust inference

844 The $p < 10^{-4}$ figures quoted in §4.5 are reported for completeness but are not the basis of the
845 claim: at tens of thousands of runs almost any non-zero slope is “significant,” and the runs are not
846 independent—they cluster within the $(N, T, \text{layout}, \text{hazard})$ configuration. The full sweep has 25,920
847 peer runs but only **81 independent design cells**. We therefore report effect sizes with cluster-robust
848 confidence intervals from a block bootstrap that resamples *configuration cells* (not rows), $B=4000$
849 resamples.

Bottleneck-shift signature: negative within-cadence correlation peaks at $K = 4$ and decays at $K = 64$

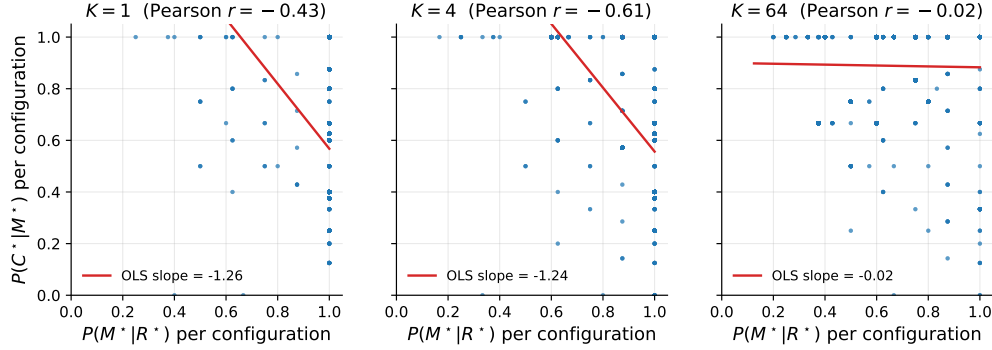


Figure 8: Per-configuration scatter of $P(M^*|R^*)$ vs. $P(C^*|M^*)$ at three cadences. The strong negative correlation at $K = 4$ (Pearson -0.61) is the bottleneck-shift signature: the same configurations that succeed at M tend to fail at C and vice versa. The correlation decays to noise by $K = 64$.

850 The two slopes of Empirical Claim 1 survive as genuine, but moderate, effects: the Spearman
851 correlation of $P(M^*|R^*)$ with K is $\rho = -0.53$ (95% cluster CI $[-0.59, -0.47]$, strong) and of
852 $P(C^*|M^*)$ with K is $\rho = +0.43$ (95% cluster CI $[+0.37, +0.50]$, moderate). Both intervals exclude
853 zero under cell-level resampling, so the bottleneck shift is not an artefact of the correlated run count;
854 it is a moderate monotone trend, and we describe it as such rather than as a near-deterministic law.

855 The interior optimum on mean time-to-success is robust in *location* but shallow in *margin*. Paired
856 block bootstrap on the cell-level t_{succ} differences gives $t_{\text{succ}}(K=8)$ below $t_{\text{succ}}(K=16)$ by 0.40 ticks
857 (CI $[0.08, 0.83]$) and below $K=64$ by 1.67 ticks (CI $[0.75, 2.83]$), both resolved; but its advantage
858 over $K=4$ (0.49, CI $[-0.47, 1.51]$) and $K=1$ (0.27, CI $[-0.43, 0.97]$) is within noise.¹ A success-
859 weighted efficiency metric (t_{succ} divided by success rate) has the same interior arg min at $K=8$, so
860 the location of the optimum is robust to the efficiency definition even though its margin over the
861 adjacent fast cadences is not resolved at this sample size. We read this as support for the *shape*
862 predicted by the $M \leftrightarrow C$ trade-off, not for a sharply tuned optimal cadence.

863 F.4 Construct validity: task success Y vs. semantic-path completion C^*

864 A success Y can in principle bypass the semantic path, so a stage bottleneck in $Q/R/M/C$ need not
865 explain variation in Y . Two facts bound this concern. First, in the multi-agent sweep that Empirical
866 Claim 1 is about, success is *defined* as reaching a `water_source` target, which is exactly the C^*
867 event; empirically Y and C^* coincide run-for-run (correlation 1.00, mean gap 0.00). The central
868 multi-agent claim is thus not exposed to a parallel-channel objection. Second, the gap $Y > C^*$ is
869 a single-agent phenomenon on the easy task families (water, rest), where a fallback controller can
870 reach the goal without a semantic target lock; it is largest exactly where success is already high and
871 the diagnosis is least needed, and it vanishes on the hard families (hazard recovery, goal-region)
872 where the framework does its work. A per-episode, stage-resolved mediation of Y (rather than of
873 C^*) requires logs that tag each success with whether it traversed the semantic path, and is a planned
874 instrumentation refinement.

875 F.5 Threshold (θ) robustness of the empty-candidate-set diagnosis

876 The single-agent attribution “candidate generation fails, not ranking” (§3.5) rests on a
877 high empty-candidate-set rate. A reviewer asks whether that is an artefact of the
878 required-tag threshold θ (deployed 0.5; the candidacy gate is `profile[tag]  $\geq \theta$` in

¹This paired bootstrap and the effect-size Spearman correlations above are computed on the same sweep execution as Table 7 (25,920 peer runs); the t_{succ} statistic is defined only over successful episodes, so the paired cell-level difference (1.67 for $K=8$ vs. $K=64$) is on that subpopulation and matches the marginal-mean difference in Table 7 (9.19–7.47=1.72) to within rounding. The effect sizes are identical on the independent `paper1_clean` execution ($-0.531/+0.434$), confirming they are not run-specific.

879 `symbolic_memory._required_matches_profile`). We sweep θ via an override on the query
880 builder (`experiments/big_experiment/theta_ablation.py`); the endpoints confirm the gate
881 is live, and the operating range shows the diagnosis is not a threshold effect.

Table 9: Empty-candidate-set rate vs. required-tag threshold θ (semantic method, assist off, hazard-recovery / goal-region). The rate is *invariant* across the entire realistic range $\theta \in [0.05, 0.95]$; only the degenerate endpoints move it, and only to the no-candidate floor. The empties are therefore absence of stored evidence, not sub-threshold filtering.

θ	empty rate (hazard recovery)	empty rate (goal region)
0.00 (accept all)	0.50	—
0.05	0.533	0.267
0.15	0.533	0.267
0.25	0.533	0.267
0.50 (deployed)	0.533	0.267
0.75	0.533	0.267
0.95	0.533	0.267
1.00 (accept none)	0.75	—

882 The reading is decisive (Table 9). At $\theta=1.0$ nothing passes and the empty rate rises; at $\theta=0.0$ the
883 required-tag gate accepts every node ($\text{conf} \geq 0$ always holds), yet the empty rate falls only to 0.50,
884 its floor—meaning that at those decision points there is *no candidate node of any confidence* to
885 retrieve. Across the entire sane range $\theta \in [0.05, 0.95]$ the rate is flat at 0.533: loosening or tightening
886 the threshold does not recover a candidate, because the required semantic place was never stored.
887 This is exactly the “absence of evidence, not contradiction” regime named in §2.3, now shown to be
888 threshold-invariant, so the memory-accumulation attribution is not an artefact of θ . (Absolute rates
889 here are on a small hazard-recovery / goal-region sweep and are not the headline 91% of §3.5, which
890 is on the full benchmark; the point is the θ -invariance, not the level.)

891 F.6 Robustness to Assumption 1 violations

892 Definition 1 rests on the conditional stage-Markov property (Assumption 1). The two practically
893 relevant violations—adaptive query rewrites not surfaced to the logger, and hidden between-stage
894 memory updates—are exactly cases where an *unlogged latent* couples non-adjacent stages. We
895 quantify the resulting estimator bias with a controlled simulation (4×10^5 episodes per setting;
896 `experiments/big_experiment/assumption1_stress_test.py`). Episodes are generated with
897 known mechanism rates $P(M^*|R^*)=0.70$, $P(C^*|M^*)=0.65$, and a hidden binary latent U that
898 shifts *both* the M-rate and the C-rate by $\pm\delta$; the naive estimator conditions only on the previous
899 starred stage, not on U .

Table 10: Bias of the naive conditional estimators under an unlogged latent of strength δ that couples the M and C stages (violating Assumption 1). $P(M^*|R^*)$ stays unbiased; $P(C^*|M^*)$ is biased upward, materially only under strong coupling.

δ (coupling strength)	bias in $\hat{P}(M^* R^*)$	bias in $\hat{P}(C^* M^*)$
0.00	−0.001	−0.002
0.05	+0.001	+0.004
0.10	+0.001	+0.015
0.15	−0.000	+0.031
0.20	+0.000	+0.056
0.25	−0.001	+0.089

900 Two conclusions (Table 10). The $R \rightarrow M$ conditional is unbiased at every δ : the latent re-weights
901 episodes but does not confound that edge. The $C \rightarrow M$ conditional acquires a one-signed *upward* bias—
902 the latent makes high-M episodes also high-C, so conditioning on M over-counts completion—but
903 the bias is only +1.5 percentage points at a mild violation ($\delta \leq 0.10$) and reaches +5.6 points only
904 under strong unlogged coupling ($\delta=0.20$). The framework’s exposure to Assumption 1 violations is
905 therefore bounded and its failure direction is known, which is the relevant robustness statement for
906 applying the logger to learned or LLM agents where such couplings are more common.

907 F.7 Preliminary continuous-substrate check (VMAS)

908 The most-requested robustness check is whether the $M \leftrightarrow C$ bottleneck shift is specific to grid sub-
 909 strates. We report a *preliminary* positive on a continuous-state substrate (VMAS (Bettini et al., 2022)).
 910 The symbolic peer memory and the operational Q/R/M/C definitions are reused unchanged; a thin
 911 discretisation bridge maps continuous (x, y) to coarse cells so the memory applies, and the controller
 912 acts in continuous space toward the materialised waypoint (experiments/vmas_portability/).
 913 Materialisation uses the same occupancy-sensitive lock as the grid planner (a candidate whose target
 914 is already claimed does not lock).

Table 11: Preliminary VMAS portability: both slope signs of Empirical Claim 1 reproduce on a continuous substrate in the coupling regime ($N=8$, $T=3$, 40 seeds, horizon 25). $\text{Spearman}(P(M^*|R^*), K) = -1.00$ and $\text{Spearman}(P(C^*|M^*), K) = +1.00$. Beyond $K \approx 32$ the system enters the independent regime (rare broadcast, no contention) and $P(M^*|R^*)$ rebounds, the same regime boundary seen on the grid.

K	$P(M^* R^*)$	$P(C^* M^*)$
1	1.000	0.182
2	0.583	0.305
4	0.448	0.405
8	0.305	0.608
16	0.278	0.639

915 We label this preliminary and do not fold it into the headline claims: it is a single scenario con-
 916 figuration, the horizon is short, there are no cluster-robust CIs yet, the memory is fed through the
 917 discretisation bridge (continuous *dynamics* with discretised symbolic *memory*, not fully continuous
 918 memory), and the reported ± 1.00 Spearman is a *perfect* rank correlation over only five monotone
 919 points ($K \leq 16$) — weak evidence on its own, indicating direction rather than a resolved effect size.
 920 What it establishes is that the two slope signs are not an artefact of grid cells: the same freshness-
 921 vs-contention mechanism operates when positions and motion are continuous. Promoting this to a
 922 headline result requires scaling to several scarcity cells and layouts with the cluster-robust bootstrap
 923 of Appendix F.3.

924 F.8 Open problems suggested by the diagnostics

925 Four concrete questions follow directly from the measurements of §3.5 and §4.5. **(1) Adaptive**
 926 **cadence.** Empirical Claim 1 assumes a fixed K ; the bottleneck-shift signature suggests an online
 927 controller K_t that broadcasts when the (estimated) M-side gain exceeds the C-side occupancy cost.
 928 **(2) Selective consolidation.** Snapshot exchange merges everything; merge rules that prioritize
 929 evidence about under-represented semantic tags attack the candidate-generation starvation of §3.5
 930 directly. **(3) Trust-weighted merging under occupancy.** The central aggregator already performs
 931 a trust-weighted merge but pays the C-collapse cost; decentralized trust rules that account for peer
 932 *intent* (not only peer evidence) could decouple freshness from target racing. *Since submission of*
 933 *the first draft, this problem has been addressed constructively:* a decentralized protocol in which
 934 an agent’s target lock piggybacks a lightweight *reservation* on its next scheduled broadcast (no
 935 coordinator, no extra channel) and locks dominated by an earlier reservation are retracted by an
 936 anti- M^* rollback with exponential backoff. On the scarcity cells this recovers the fast-cadence
 937 completion loss on *unconditional* metrics — success rate rises at every $K \in \{1, 2, 4\}$ with disjoint
 938 bootstrap CIs, and fast sharing with the protocol exceeds the $K=8$ optimum baseline (0.614–0.620 vs.
 939 0.583) — with rollback latency bounded by $\approx K$ ticks. Details in a companion manuscript; the open
 940 problem stands revised toward its harder variants (adaptive reservation scope, heterogeneous trust).
 941 **(4) Substrate generality.** The decomposition transfers across MiniGrid, BabyAI, the custom grid,
 942 and a standard-MiniGrid multi-agent wrapper with identical operational definitions and verified slope
 943 signs of Empirical Claim 1 (§ 5), and a preliminary check reproduces both slopes on the continuous-
 944 state VMAS substrate (Appendix F.7). A full continuous-state validation with cluster-robust CIs, and
 945 richer 3D substrates (Habitat (Savva et al., 2019)), remain the next portability steps.

F.9 Extended related work

Classical cognitive-map perspectives motivate explicit place abstractions (Tolman, 1948; O’Keefe and Nadel, 1978; Kuipers, 2000); modern embodied AI combines mapping and planning (Gupta et al., 2017) or exposes topological memory (SPTM, Savinov et al., 2018); goal-conditioned RL conditions on explicit goals or embeddings (Andrychowicz et al., 2017; Ghosh et al., 2021), with successor-representation work providing predictive structure (Dayan, 1993; Stachenfeld et al., 2017; Momennejad et al., 2017). Our setting differs in that the target is a semantic pattern retrieved from memory rather than a coordinate. In multi-agent RL, CTDE dominates (Lowe et al., 2017; Foerster et al., 2018) and learned communication adds a per-step exchange variable (Sukhbaatar et al., 2016; Foerster et al., 2016); gossip and consensus protocols (Boyd et al., 2006; Xiao et al., 2007) are the closest network-level analogue of our cadence axis, lifted here to the cognitive-architecture level. LLM-agent systems combine reasoning/acting, reflection, explicit memory, and multi-agent conversation (Yao et al., 2023; Shinn et al., 2023; Wang et al., 2023; Park et al., 2023; Wu et al., 2023; Packer et al., 2023; Chhikara et al., 2025). Our framework is orthogonal: rather than learning a communication policy end-to-end, it *diagnoses* where in the Q/R/M/C chain a given architecture succeeds or fails, and yields a slope-level prediction about cadence that is testable on logs.

F.10 SOTA baseline and coverage matrix

Table 13 separates direct experimental baselines from broader SOTA references. The local experiments are intentionally modest and fully instrumented: random / graph-only / SPTM-like / learned BC in MiniGrid, CommNet PPO in the multi-agent grid, and a TextNav LLM step-budget study over two local Ollama models (in which Q/R/M/C localizes a completion-limited failure, §4.5). ReAct, Reflexion, Voyager, Generative Agents, AutoGen, MemGPT, and Mem0 are not reimplemented as full systems here; they define the current memory-agent design space against which Q/R/M/C is positioned as a diagnostic protocol.

Positioning against SOTA memory designs. The memory-agent design space can be organised on two axes: single- vs. multi-agent, and how memory is represented and shared (Table 12). Vector-embedding memory (MemGPT, Voyager single-agent; Generative Agents multi-agent) retrieves by opaque nearest-neighbour lookup, so a failure is not localisable to a stage. Shared-context designs (MetaGPT, AutoGen) pool everything into one window, dissolving per-agent state. Learned-communication MARL (CommNet, and stronger CTDE methods QMIX (Rashid et al., 2018), MAPPO (Yu et al., 2022), MADDPG (Lowe et al., 2017)) optimises a message channel end-to-end and, as we verify for CommNet PPO (§4.5), emits no separable query/lock/completion events. The cell our system occupies—per-agent *symbolic* memory with an explicit query/retrieve/materialise interface, exchanged at a tunable cadence—is empty in this matrix, and it is exactly the interface that makes the four Q/R/M/C stages observable. Our claim against SOTA is therefore not higher reward but higher *diagnosability at matched task success*: the same interface that a strong learned method trades away for end-to-end optimisation is what lets the framework localise where memory helps.

Table 12: Where the framework sits in the memory-agent design space. Q/R/M/C stage-observability requires an explicit query/lock interface, which the symbolic-per-agent cell provides.

Memory representation	Single-agent	Multi-agent
Vector-embedding memory	MemGPT, Voyager (Packer et al., 2023; Wang et al., 2023)	Generative Agents (Park et al., 2023)
Shared context window	—	AutoGen (Wu et al., 2023)
Learned-comm. MARL	—	CommNet, QMIX, MAPPO, MADDPG (Sukhbaatar et al., 2016; Rashid et al., 2018; Yu et al., 2022; Lowe et al., 2017)
Symbolic per-agent + explicit interface	Q/R/M/C single-agent (§3)	this work (peer memory, cadence K)

983 **F.11 Multi-agent oracle-intervention numerics**

984 **F.12 Oracle C: definition and reading**

985 Oracle C is the completion-stage analogue of oracle R and oracle M, added to close the R/M/C triple
986 symmetrically. The real Q/R/M pipeline runs unchanged (real memory query, real materialisation),
987 and once an agent has locked a target that is a genuine `water_source`, the oracle guarantees
988 arrival by placing the agent on that cell; scarcity is preserved because a cell already claimed or
989 occupied is not available, so oracle C cannot make more than T agents succeed. The runner is
990 `experiments/big_experiment/exp_oracle_interventions.py` (mode `oracle_C`).

991 Two readings follow from Table 14. (i) Oracle C produces the *lowest* mean t_{succ} of the three
992 interventions (~ 6.1 – 6.4 vs. ~ 7.2 for R and M), because it removes the entire post-lock travel
993 time; this confirms that a real part of the excess time is controller travel to a correctly materialised
994 target. (ii) Oracle C yields the *smallest* success-rate gain ($+0.01$ – 0.02 over baseline, below both
995 oracle R at 0.614 and oracle M at 0.602), because in scarcity the constraint on *whether* an agent
996 succeeds is upstream—retrieving and locking a still-available target—not reaching it. The two facts
997 together are exactly the framework’s thesis in intervention form: completion is where time is spent,
998 retrieval/materialisation is where success is won.

999 **F.13 Stage-agnostic cadence hypotheses and what the framework changes**

1000 A first version of the multi-agent cadence sweep (12,960 runs at 20 seeds; subsumed by the 35,640-run
1001 sweep in this paper) was originally planned with four operational hypotheses formulated *before*
1002 Empirical Claim 1. They are retained here because they motivated the eventual framework. The
1003 framework was formulated after the sweep was complete; it is not claimed to have predicted the
1004 sweep numbers a priori (see “Chronology” in §4.3).

1005 **Original hypotheses (pre-framework).**

1006 **H1.** An interior $K^* \in \{2, 4, 8\}$ minimising mean time-to-success exists for most (N, T, layout)
1007 cells.

1008 **H2.** K^* grows monotonically with scarcity $\rho = N/T$.

1009 **H3.** When $\rho \leq 1$, $K^* = 1$.

1010 **H4.** Peer-broadcast architectures Pareto-dominate centralized aggregation on (mean time-to-success,
1011 success rate).

1012 **Stand-alone result on the original 20-seed sweep.** The practical claim was directionally supported:
1013 peer at the empirically best K was faster than centralized in the scarcity regime (a modest effect;
1014 Mann–Whitney $p=0.044$ on this small standalone sweep, reported as exploratory rather than as
1015 primary evidence, consistent with the effect-size-first line of Appendix F.3). Among the four
1016 structural hypotheses, only H4 was clearly supported in the random-layout subset. H1 held in 7 of
1017 18 cells; H2 had a non-significant negative correlation; H3 was supported in 4 of 9 admissible cells.
1018 The 35,640-run extended sweep used elsewhere in the paper produces matching stand-alone numbers
1019 within bootstrap uncertainty.

1020 **What the framework changes.** The hypotheses H1–H4 are stage-agnostic: each ties K^* to scarcity
1021 ρ without specifying mechanism. Empirical Claim 1, formulated after the sweep, ties the slopes of
1022 $P(M^*|R^*)$ and $P(C^*|M^*)$ to cadence; the slope conditions are testable on logs. The data verify
1023 both signs as moderate, cluster-robust effect sizes (Appendix F.3). We make no formal claim about
1024 the shape of the conditional product Φ , on which the data are monotone. The interior optimum holds
1025 on the efficiency metric mean time-to-success.

1026 **G Assist on/off analysis**

1027 Across all four single-agent task families, top-line success is essentially unchanged under assist on/off
1028 ($\Delta \leq 0.10$). The only non-trivial effect is on rest, where assists improve post-retrieval completion
1029 by $+0.05$ without changing query/retrieval rates, consistent with assists acting on materialization and
1030 post-retrieval execution.

1031 H Retrained retriever and CommNet baseline

1032 H.1 Retrained candidate-generation MLP (§3.5)

1033 **Dataset extraction.** From the 10-seed hazard-recovery benchmark (MiniGrid-LavaGapS7-v0,
1034 method `milestone_state_conditioned_hazard_recovery_v7`), an 18-dimensional feature
1035 vector was extracted for every step of every episode of every seed — 3 numeric agent-state scalars
1036 (energy, thirst, risk_budget), 9 semantic-tag confidences, 6 intent one-hot — and a binary label: *was
1037 this symbolic node ever selected as a candidate during the run?* Dataset totals: 814 examples, 113
1038 positives (13.9%).

1039 **Split.** By seed: train {2, 3, 4, 5, 6, 7} (505 examples, 9.7% positive), val {8, 9} (170, 22.4% positive),
1040 test {10, 11} (139, 18.7% positive).

1041 **Architecture.** $18 \rightarrow 32 \rightarrow 32 \rightarrow 1$, ReLU activations, BCE with $\text{pos_weight} \approx 9.3$ for class
1042 imbalance, Adam $\text{lr} = 2 \cdot 10^{-3}$ with $\text{weight_decay} = 10^{-4}$, batch 64, 200 epochs, best by val loss.

1043 Results.

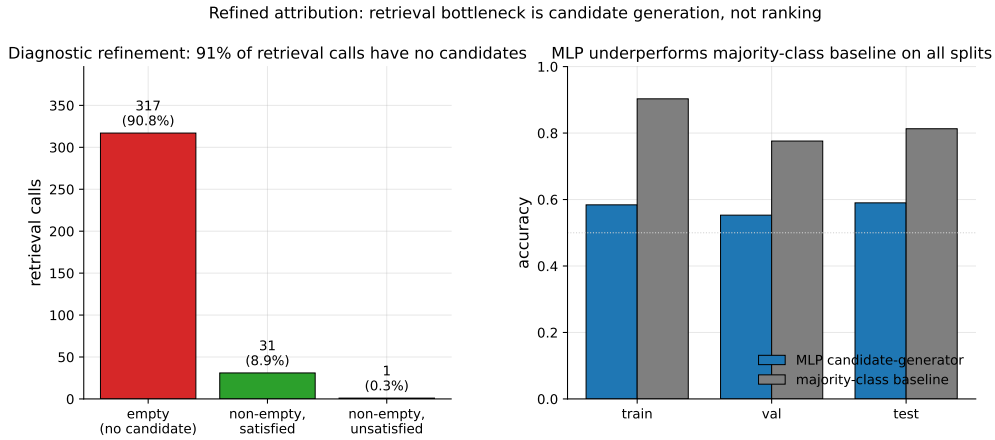


Figure 9: Refined attribution diagnostic. *Left*: 349 retrieval calls aggregated across 10 seeds; 317 (90.8%) returned empty candidate sets, 31 (8.9%) returned at least one satisfied candidate, only 1 (0.3%) returned candidates that were ranked but unsatisfied. The bottleneck is candidate generation, not candidate ranking. *Right*: the trained MLP underperforms a majority-class baseline on all three seed-disjoint splits, confirming that instantaneous state features alone are insufficient to predict candidate eligibility.

1044 The MLP underperforms majority-class on accuracy across all splits and achieves only moderate
1045 precision at fixed recall. Instantaneous state features are insufficient to predict candidate eligibility
1046 downstream. This is the negative-but-informative finding referenced in §3.5: the framework’s
1047 “retrieval-limited” attribution refines to “memory-accumulation-limited” rather than “ranking-limited”.

1048 H.2 CommNet-style learned communication baseline (§4.5)

1049 **Policy architecture.** Per-agent observation is 4-dim direction one-hot + $5 \times 5 \times 5$ local cell-type window
1050 (125) + 2-dim normalised position = 131. Shared-weights encoder $131 \rightarrow 64 \rightarrow 64$ (ReLU), 16-dim
1051 comm projection, peer mean-pool (excluding self), combine $80 \rightarrow 64$ (ReLU), actor head $64 \rightarrow 4$
1052 (TURN_LEFT, TURN_RIGHT, FORWARD, NOOP), critic head $64 \rightarrow 1$. This is a CommNet-style
1053 learned-communication architecture (Sukhbaatar et al., 2016).

1054 **Training.** REINFORCE (Williams, 1992) with value-baseline; loss = $-\text{mean}(\log \pi(a) \cdot (R - V)) +$
1055 $0.5 \cdot \text{MSE}(V, R) - 0.01 \cdot H[\pi]$; Adam $\text{lr} = 3 \cdot 10^{-4}$, gradient clipping 1.0. Reward shaping: +1.0
1056 on first success per agent, -0.01 step cost, -0.1 per hazard contact. 2000 episodes cycling 200
1057 environment seeds, wall-clock 100 s on CPU. Held-out eval: 50 environment seeds 250–299.

1058 **Training curve.** Episode-level success (last-100 window) stays in $[0.13, 0.23]$ throughout training;
 1059 no monotonic improvement after ~ 200 episodes. The policy converges to a local optimum where
 1060 one of three agents finds water by chance.

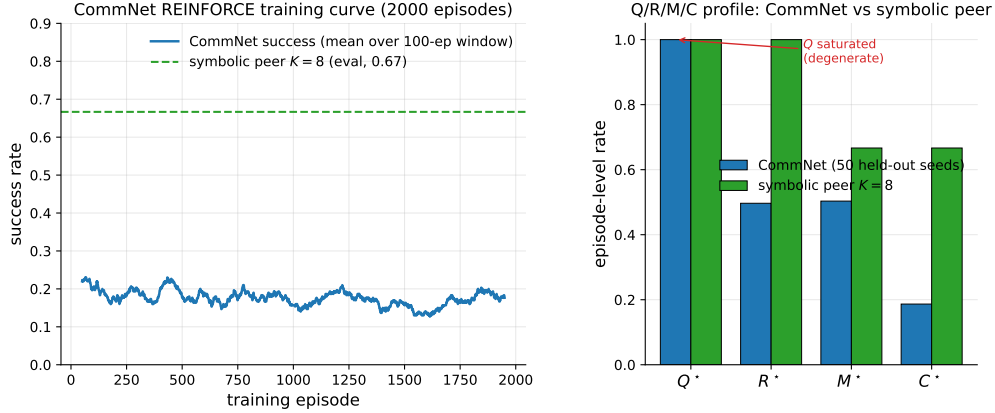


Figure 10: Learned-communication baseline diagnostics. *Left:* CommNet training curve (smoothed success rate over a 100-episode window) over 2000 training episodes; success stays below 0.25 throughout. The dashed line is the symbolic peer architecture at $K=8$ on the same scenario for reference. *Right:* Q/R/M/C profile of the trained CommNet on 50 held-out seeds vs symbolic peer at $K=8$. Q^* is saturated at 1.0 on CommNet (degenerate by construction); R/M collapse together; only on the symbolic architecture are the conditional rates separated.

1061 **Q/R/M/C evaluation (50 held-out seeds, stochastic action sampling).**

1062 Compare to the symbolic peer architecture at $K=8$: $P(M^*|R^*)=0.69$, $P(C^*|M^*)=0.87$, success
 1063 ~ 0.65 .

1064 **PPO variant: structural degeneracy verified at competitive success.** A PPO-trained (Schulman et al., 2017) variant of the same CommNet architecture (clip $\epsilon=0.2$, generalized advantage estimation (Schulman et al., 2016) with $\lambda=0.95$, $\gamma=0.99$, 4 PPO epochs per rollout, 64 rollouts per update, 200 updates, ≈ 96 minutes on a single CPU core; full setup in experiments/commnet_ppo_baseline/train_ppo_commnet.py) reaches 0.667 success on 100 held-out seeds — competitive with the symbolic peer architecture at $K=8$ (0.65).

1070 The PPO result converts the structural argument from an assertion (“training budget would not change this”) to a measurement: Q^* remains saturated at 1.00, $P(M^*|R^*)$ remains 1.00 (R/M collapse), even
 1071 at competitive success. The Q stage is degenerate by construction in any policy whose communication
 1072 channel is a real-valued vector with a learned linear projection — the channel is never informatively
 1073 empty, regardless of training budget — and without an explicit symbolic target-lock interface the M
 1074 event cannot separate from R. Only architectures with an explicit *query/lock* interface (the symbolic
 1075 stack of §4.1) expose stage-separable Q/R/M/C events.
 1076

1077 I Reproducibility commands

1078 An anonymized public repository accompanies this submission. The main paper artifacts were
 1079 generated from the following scripts; PYTHONPATH=. and a Python 3 virtual environment with numpy,
 1080 scipy, pandas, matplotlib, imageio, gymnasium, and minigrid are assumed throughout.

1081 **One-command reproduction (multi-agent block, ≈ 22 minutes on 8 cores)**

1082 bashscripts/reproduce_paper.sh

1083 Clean Paper 1 suite

- 1084 • smoke suite (all local non-LLM blocks): `pythonscripts/run_paper1_clean_experiments.py --modesmoke --skip_llm --out_dir/tmp/paper1_clean_experiments_smoke`
- 1085
- 1086 • full local suite (single-agent, oracle, multi-agent, CommNet, portability/noise):
1087 `pythonscripts/run_paper1_clean_experiments.py --modefull --skip_`
1088 `llm --workers8 --out_dir/tmp/paper1_clean_experiments_full`
- 1089 • LLM TextNav step-budget sweep (requires local Ollama), run once per budget $K' \in \{6, 12, 25\}$:
1090 `python-mexperiments.llm_collective.run_model_sweep --modelsqwen3:4bllama3.`
1091 `1:latest --episodes8 --step_limit<K'> --out_dir/tmp/llm_steplimit_<K'>`

1092 Single-agent (§3.2–§3.4)

- 1093 • article benchmark and learned-baseline suite: `scripts/benchmark_symbolic_memory_article.py`
- 1094
- 1095 • article analysis and figure generation: `scripts/analyze_symbolic_memory_article.py`
- 1096 • synthetic factorization sanity check: `scripts/validate_qrmc_factorization.py`
- 1097 • oracle-stage intervention benchmark: `scripts/benchmark_oracle_stage_interventions.py`
- 1098
- 1099 • BabyAI transfer benchmark: `scripts/compare_semnav_babyai.py`

1100 Multi-agent (§4.5)

- 1101 • Base batch — peer at $K \in \{1, 2, 4, 8, 16\}$ and the three cadence-free architectures (in-
1102 dependent, shared bus, central aggregator), 40 seeds (25,920 runs = 16,200 peer + 9,720
1103 cadence-free), ≈ 18 min: `pythonexperiments/big_experiment/exp_cadence_phase.py --modefull --workers8 --out_dir/tmp/big_experiment_qrmc_40`
- 1104
- 1105 • Extended-cadence batch — peer at $K \in \{32, 48, 64\}$ only (9,720 runs), ≈ 4 min:
1106 `pythonexperiments/big_experiment/exp_extra_K.py --workers8 --out_dir/tmp/`
1107 `big_experiment_extraK`
- 1108 *Note.* These two batches partition the 35,640 total by *cadence range*; the architecture-wise
1109 partition of Appendix F.2 (25,920 peer over all eight cadences vs. 9,720 cadence-free) reuses
1110 the same two totals for *different* subsets (a coincidence of 3 extended cadences $\times$ 3,240 = 3
1111 cadence-free architectures $\times$ 3,240). The combined log is `tmp/paper1_clean_experiments_`
1112 `full/multiagent_cadence_full/runs.csv`.
- 1113 • Q/R/M/C stage analysis and bootstrap CIs: `pythonexperiments/big_experiment/`
1114 `analyze_qrmc.py --runs_csvtmp/big_experiment_qrmc_40/runs.csv --out_`
1115 `dirtmp/big_experiment_qrmc_40`
- 1116 • Multi-agent oracle interventions: `pythonexperiments/big_experiment/exp_oracle_`
1117 `interventions.py --workers8 --out_dir/tmp/big_experiment_oracle`
- 1118 • Phase 1–4 smoke tests: `pythonscripts/run_all_smokes.py --out_dir/tmp/all_smokes`
- 1119 • PPO-trained CommNet baseline (Table 15, ≈ 96 min on a single CPU core):
1120 `python-mexperiments.commnet_ppo_baseline.train_ppo_commnet --total_`
1121 `updates200 --rollouts_per_update64`
- 1122 • Q/R/M/C eval of the PPO policy on 100 held-out seeds: `pythonexperiments/commnet_`
1123 `baseline/eval_with_qrmc.py --policy_pathexperiments/commnet_ppo_baseline/`
1124 `ppo_policy.pt --out_dir/experiments/commnet_ppo_baseline --n_episodes100`
- 1125 • MiniGrid-substrate portability sweep (§5, 100 runs, ≈ 2 min): `python-mexperiments.`
1126 `minigrid_multiagent_wrapper.run_portability_sweep --out_dir/tmp/minigrid_`
1127 `portability`

1128 J Asset and license note

1129 The experiments use MiniGrid (Chevalier-Boisvert et al., 2023) and BabyAI (Chevalier-Boisvert
1130 et al., 2018), both released under the MIT License. The locally implemented baselines and benchmark

1131 runner scripts introduced in this paper will be released under the MIT License as part of the camera-
1132 ready artifact package.

Table 13: SOTA coverage for Paper 1. “Instrumented” means the run emits the operational Q^* , R^* , M^* , C^* events used by Definition 1.

Family	Representative method	Status in this paper	Q/R/M/C comparison point
Topological navigation memory	SPTM-like graph baseline (Savinov et al., 2018)	Direct MiniGrid baseline	Tests whether topological memory without semantic query/lock exposes non-trivial stage events. Strong success on easy tasks but no explicit query/lock interface, so diagnostics are structurally limited.
Behavior cloning	Learned grid-observation BC	Direct MiniGrid baseline	
Learned communication	CommNet PPO (Sukhbaatar et al., 2016; Schulman et al., 2017)	Direct multi-agent baseline	Competitive success; Q^* saturates and R/M collapse, verifying observability loss.
Cooperative MARL (CTDE / value decomposition)	QMIX, MAPPO, MADDPG (Rashid et al., 2018; Yu et al., 2022; Lowe et al., 2017)	Primary-source reference	Stronger end-to-end baselines than CommNet; they optimise a joint policy/message channel without an explicit query/lock interface, so the same observability-collapse argument applies — the framework quantifies the diagnostic cost of implicit coordination rather than competing on reward.
Reasoning/acting LLM agents	ReAct (Yao et al., 2023)	Primary-source reference	Q/R/M/C can instrument the resulting action traces if queries, retrieved evidence, target locks, and completions are logged.
Reflective LLM agents	Reflexion (Shinn et al., 2023)	Primary-source reference	Reflection can be treated as upstream Q/R revision; the framework asks whether it improves R, M, or C.
Embodied LLM agents	Voyager (Wang et al., 2023)	Primary-source reference	Skill/memory retrieval maps naturally onto R and M, but full reproduction is outside this paper.
Social/memory simulations	Generative Agents (Park et al., 2023)	Primary-source reference	Their observation/memory/reflection loop motivates logging stage-specific memory use rather than only believability.
Multi-agent LLM frameworks	AutoGen (Wu et al., 2023)	Primary-source reference	Conversation traces provide candidate Q/R/M/C events; this paper supplies the measurement contract.
Long-term memory systems	MemGPT / Mem0 (Packer et al., 2023; Chhikara et al., 2025)	Primary-source reference	Persistent memory systems can be evaluated by where they fail: retrieval, materialization, or completion.
Local LLM sub-strate	qwen3:4b, llama3.1	Direct TextNav step-budget study	Q/R/M/C localizes a completion-limited failure ($C^* \rightarrow 0$ while $Q/R/M$ stay saturated) in real LLM agents; see Table 3.

Table 14: Multi-agent oracle interventions on the scarcity cases at three cadences. Mean time-to-success (and success rate) averaged across $(N, T) \in \{(3, 2), (5, 3), (8, 5)\}$, two layouts (asymmetric, random), 15 seeds. Oracle R returns ground-truth water positions; oracle M locks the nearest real water; **oracle C** guarantees completion of a real materialised lock (Appendix F.12). Oracle C gives the largest *time* reduction (it removes travel) but the smallest *success* gain — completion is a time cost, not the success bottleneck, which sits upstream at R/M.

K	Baseline	Oracle R	Oracle M	Oracle C
<i>mean t_{succ}</i>				
1	10.33	7.22	7.21	6.19
4	10.98	7.22	7.21	6.07
16	8.45	7.22	7.21	6.41
<i>success rate</i>				
1	0.591	0.614	0.602	0.591
4	0.575	0.614	0.602	0.600
16	0.578	0.614	0.602	0.598

Metric	Train	Val	Test
Accuracy	0.584	0.553	0.590
Majority-class baseline accuracy	0.903	0.776	0.813
Precision @ recall = 0.70	0.171	0.278	0.275

Quantity	Value
Q^* rate	1.00 (saturated)
R^* rate	0.50
M^* rate	0.50 (collapses onto R)
C^* rate	0.19
$P(R Q)$	0.50
$P(M R)$	1.00
$P(C M)$	0.42
Success rate	0.19
Mean steps	80 (budget exhausted)

Table 15: REINFORCE vs PPO CommNet vs symbolic peer ($K=8$) on the scarcity scenario ($N=3$, $T=2$, asymmetric, hazard 0.05). PPO is competitive on success rate, yet still exhibits Q-saturation and R/M collapse — the diagnostic claim verified, not asserted.

Metric	CommNet REINFORCE	CommNet PPO	Symbolic peer ($K=8$)
Success rate	0.19	0.667	0.65
Q^* rate	1.00	1.00	0.99
R^* rate	0.50	0.997	0.69
M^* rate	0.50	0.997	0.69
$P(M^* R^*)$	1.00	1.00	0.69
$P(C^* M^*)$	0.42	0.67	0.87
R/M separated?	no	no	yes

1133 NeurIPS Paper Checklist

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1138 Justification: The abstract and introduction present the paper primarily as a four-stage
1139 diagnostic framework with a reference symbolic-semantic instantiation, and they explicitly
1140 scope the evidence to benchmarked navigation environments with a limited portability check
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1252 An anonymized public release of the codebase — single-agent benchmark, multi-agent grid
1253 environment, 35,640-run sweep driver, Q/R/M/C analyser, oracle-intervention scripts, and
1254 exact commands to reproduce every table and figure — accompanies the submission as
1255 supplemental material (anonymized GitHub repository; single-command reproduction of
1256 the main multi-agent sweep on an 8-core machine in ≈ 22 minutes, matching Appendix H).

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Justification: The paper reports 95% bootstrap confidence intervals over seed-level aggregates ($B=5,000$ resamples; Section 3.1, “Protocol” paragraph) and uses paired Wilcoxon signed-rank tests with Holm-Bonferroni correction for assist comparisons. CIs are shown in Tables 3 and 5 (Best semantic per-task rates; the 35,640-run MOP column) and in Figures 4–6 (bootstrap shaded bands).

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1427 13. New assets

1428 Question: Are new assets introduced in the paper well documented and is the documentation
1429 provided alongside the assets?

1430 Answer: [\[No\]](#)

1431 Justification: The submission introduces benchmark scripts and analysis artifacts. A formal
1432 anonymized release package with documentation is planned for camera-ready; Appendix H
1433 lists all scripts needed to reproduce each reported result, and Appendix I states the intended
1434 license for the new assets.

1435 Guidelines:

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- 1437 • Researchers should communicate the details of the dataset/code/model as part of their
1438 submissions via structured templates. This includes details about training, license,
1439 limitations, etc.
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1441 asset is used.
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1443 create an anonymized URL or include an anonymized zip file.

1444 14. Crowdsourcing and research with human subjects

1445 Question: For crowdsourcing experiments and research with human subjects, does the paper
1446 include the full text of instructions given to participants and screenshots, if applicable, as
1447 well as details about compensation (if any)?

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1449 Justification: The paper does not involve crowdsourcing or human-subject experiments.

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